



Occurrence and effects of plastic additives on marine environments and organisms: a review

Ludovic Hermabessiere, Alexandre Dehaut, Ika Paul-Pont, Camille Lacroix, Ronan Jézéquel, Philippe Soudant, Guillaume Duflos

► To cite this version:

Ludovic Hermabessiere, Alexandre Dehaut, Ika Paul-Pont, Camille Lacroix, Ronan Jézéquel, et al.. Occurrence and effects of plastic additives on marine environments and organisms: a review. *Chemosphere*, 2017, 182, pp.781-793. <10.1016/j.chemosphere.2017.05.096>. <anses-02556379>

HAL Id: anses-02556379

<https://anses.hal.science/anses-02556379v1>

Submitted on 28 Apr 2020

HAL is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



HAL Authorization

Occurrence and effects of plastic additives on marine environments and organisms: a review

Ludovic HERMABESSIERE^a, Alexandre DEHAUT^a, Ika PAUL-PONT^b, Camille LACROIX^c, Ronan JEZEQUEL^c, Philippe SOUDANT^b & Guillaume DUFLOS^{a*}

^a Anses, Laboratoire de Sécurité des Aliments, Boulevard du Bassin Napoléon, 62200 Boulogne-sur-Mer, France

^b Laboratoire des Sciences de l'Environnement Marin (LEMAR), UMR6539/UBO/CNRS/IRD/IFREMER, Institut Universitaire Européen de la Mer, Technopôle Brest-Iroise, rue Dumont d'Urville, 29280 Plouzané, France

^c CEDRE, 715 rue Alain Colas, 29218 Brest Cedex 2, France

* Corresponding author:

guillaume.duflos@anses.fr

Phone: + 33 3 21 99 25 00

Abstract

Plastics debris, especially microplastics, have been found worldwide in all marine compartments. Much research has been carried out on adsorbed pollutants on plastic pieces and hydrophobic organic compounds (HOC) associated with microplastics. However, only a few studies have focused on plastic additives. These chemicals are incorporated into plastics from which they can leach out as most of them are not chemically bound. As a consequence of plastic accumulation and fragmentation in oceans, plastic additives could represent an increasing ecotoxicological risk for marine organisms. The present work reviewed the main class of plastic additives identified in the literature, their occurrence in the marine environment, as well as their effects on and transfers to marine organisms. This work identified polybrominated diphenyl ethers (PBDE), phthalates, nonylphenols (NP), bisphenol A (BPA) and antioxidants as the most common plastic additives found in marine environments. Moreover, transfer of these plastic additives to marine organisms has been demonstrated both in laboratory and field studies. Upcoming research focusing on the toxicity of microplastics should include these plastic additives as potential hazards for marine organisms, and a greater focus on the transport and fate of plastic additives is now required considering that these chemicals may easily leach out from plastics.

Keywords

Microplastics, Plastic additives, Bisphenol A, Phthalates, Brominated Flame retardant

1. Introduction

Due to their numerous societal benefits, plastics hold an important place in human society (Andrady and Neal, 2009). Plastic, a man-made material, is inexpensive, strong, durable, lightweight and easy to produce (Thompson *et al.*, 2009). As a consequence, plastic production has been increasing since the 1950s, and notably rose from 225 million tons in 2004 to 322 million tons in 2015, representing a 43% increase over the last decade (PlasticsEurope, 2016). However, this estimate does not take into account the proportion of synthetic fibers which are widely used in the textile and fishery industries (Dris *et al.*, 2016) and there is an underestimation of 15% to 20% depending on the year (Industrievereinigung Chemiefaser, 2013). Low estimates predicted that floating marine plastic weigh between 70,000 and 270,000 tons (Cózar *et al.*, 2014; Eriksen *et al.*, 2014; Van Sebille *et al.*, 2015). Small pieces of plastics called microplastics (MP) account for a total of 51 trillion plastic debris (Van Sebille *et al.*, 2015).

Microplastics have been defined as plastics particles smaller than 5 mm (Arthur *et al.*, 2009). Growing attention has been accorded to microplastics during the last decade, since the publication by Thompson *et al.* (2004). Micro-sized plastic pieces originate from two distinct pathways: primary and secondary sources. Primary sources of MP correspond to (i) plastics that are directly manufactured at micrometric size, including plastic pellets (Barnes *et al.*, 2009; Cole *et al.*, 2011), (ii) MP from exfoliating cosmetics (Chang, 2015; Fendall and Sewell, 2009; Napper *et al.*, 2015; Zitko and Hanlon, 1991) and (iii) clothing fibers found in wastewater treatment plants (Browne *et al.*, 2011; Carr *et al.*, 2016). Secondary MP results from the breakdown of larger pieces due to mechanical abrasion and photochemical oxidation in the environment (Andrady, 2011; Bouwmeester *et al.*, 2015; Lambert and Wagner, 2016). MP can also degrade into smaller pieces called nanoplastics (Gigault *et al.*, 2016; Koelmans *et al.*, 2015; Lambert and Wagner, 2016).

Due to their small size, MP can be ingested by a wide range of marine organisms such as zooplankton, bivalves and worms (De Witte *et al.*, 2014; Devriese *et al.*, 2015; Graham and Thompson, 2009; Rochman *et al.*, 2015; Sussarellu *et al.*, 2016; Van Cauwenberghe and Janssen, 2014; Van Cauwenberghe *et al.*, 2015) and by organisms from higher trophic levels such as fish (Boerger *et al.*, 2010; Carpenter *et al.*, 1972; Dantas *et al.*, 2012; Foekema *et al.*, 2013; Lusher *et al.*, 2013; Neves *et al.*, 2015; Possatto *et al.*, 2011; Rochman *et al.*, 2015) and marine mammals (Eriksson and Burton, 2003; Lusher *et al.*, 2015). This ingestion of MPs can result in physical damage such as obstruction or internal abrasions (Wright *et al.*, 2013). In addition to these physical threats, MP can potentially transfer chemicals adsorbed on their surface (Mato *et al.*, 2001; Teuten *et al.*, 2007; Teuten *et al.*, 2009) or plastic additives. However, less attention has been paid to the transfer of plastic additives to marine organisms in comparison with hydrophobic organic compounds (HOC), despite the fact that many additives have been recognized as hazardous (Lithner *et al.*, 2011). Therefore, the transport and fate of plastic additives leaching out from plastic debris should definitely be carefully addressed in future field, laboratory and modelling works.

Plastics are made by polymerizing monomers and other substances (Lithner *et al.*, 2011) including plastic additives. Plastic additives are chemical compounds, like plasticizers, which provide required properties to a plastic polymer or are incorporated to facilitate the manufacturing process (OECD, 2004). Moreover, some plastic additives are used as monomers, for example bisphenol A is the monomer of polycarbonate (PC) but also a stabilizer in other polymers. Plastic additives are mainly used as plasticizers, flame retardants, stabilizers, antioxidants and pigments. Phthalates, BPA, nonylphenols, and brominated flame retardants (BFR) are the most common additives recovered from the environment (Bergé *et al.*, 2012; David *et al.*, 2009; de Boer *et al.*, 1998; de los Ríos *et al.*, 2012; Mackintosh *et al.*, 2004; Net *et al.*, 2015; Xie *et al.*, 2005; Xie *et al.*, 2007) and represent a hazard to the environment and

organisms (Lithner *et al.*, 2011; Meeker *et al.*, 2009; Oehlmann *et al.*, 2009). These plastic additives are released into the marine environment by numerous pathways including industrial and municipal wastewater, atmospheric deposition, runoff and river transport resulting from application of sewage sludge in agriculture. In addition leaching of plastic additives from macro and microplastics is known to occur in the marine environment. Thus, the accumulation and degradation of plastic debris might represent another major input of these chemical compounds in oceans. As a consequence, more research is needed on the hazards of plastic additives associated with microplastics.

The aim of this review is to (i) list and describe the most predominant plastic additives used worldwide in the plastic industry, (ii) present an overview of the occurrence of plastics additives in the marine environment, and (iii) document the effects of plastic additives on marine organisms. Lastly, recommendations will be made in order to identify the polymer-additives pairs of major concern on which further research should focus.

2. Chemicals used as plastic additives

Multiple types and families of chemicals are mixed with polymers to produce plastics. The type of additive depends on the plastic polymer and the requirements of the final product (Table 1).

102 **Table 1: List of the most commonly produced polymers and their associated plastic additives. Adapted from Hansen *et al.* (2013)**

Polymer	Consumption in the EU27 (in million tons) in 2015 ¹	Additive types	Amount in polymers (% w/w)	Hazardous substances ²
PP	9	Antioxidant	0.05 – 3	Bisphenol A; Octylphenol; Nonylphenol
		Flame retardant (cable insulation and electronic applications)	12 – 18	Brominated flame retardant; Boric acid; Tris(2-chloroethyl)phosphate
HDPE	8	Antioxidant	0.05 - 3	Bisphenol A; Octylphenol; Nonylphenol
		Flame retardant (cable insulation application)	12 -18	Brominated flame retardant; Boric acid; Tris(2-chloroethyl)phosphate
LDPE	6	Antioxidant	0.05 – 3	Bisphenol A; Octylphenol; Nonylphenol
		Flame retardant (cable insulation application)	12 – 18	Brominated flame retardant; Boric acid; Tris(2-chloroethyl)phosphate
PVC	5	Plasticizer	10 – 70	Phthalate
		Stabilizer	0.5 – 3	Bisphenol A; Nonylphenol
PUR	3.5	Flame retardant	12 - 18	Brominated flame retardant; Boric acid; Tris(2-chloroethyl)phosphate

¹ According to [PlasticsEurope \(2016\)](#); PP: Polypropylene; HDPE: High Density Polyethylene; LDPE: Low Density Polyethylene; PVC: Polyvinyl Chloride; PUR: Polyurethane.

² Hazardous substances refer to chemicals that pose a risk to the environment and to human health as defined by the REACH regulation in the European Union according to the [European Chemical Agency \(2017\)](#).

The following section describes the most common additive types used in the manufacturing processes (Table 2) that have been reported in macro- and microplastic debris collected in environmental surveys: brominated flame retardants, phthalates used as plasticizers, nonylphenols, bisphenol A and antioxidants.

Table 2: List of common plastic additives and their associated functions and potential effects

Additives	Function	Effects
Brominated Flame Retardants (BFR)	Reduce flammability in plastic. Also adsorbed on plastic from the surrounding environment.	Potential endocrine disruptors
Phthalates	Plasticizers to soften plastic mainly in polyvinyl chloride.	Endocrine disruptors
Nonylphenol	Antioxidant and plasticizer in some plastics	Endocrine disruptors
Bisphenol A (BPA)	Monomer in polycarbonate and epoxy resins. Antioxidant in some plastics.	Endocrine disruptors Estrogen mimic
Irganox®	Antioxidant in some plastics.	

The main plastic additives described are listed in Table 3 with their associated octanol-water partition coefficient (K_{ow}). K_{ow} has been used for predicting how a chemical will concentrate in marine organisms and an increase in $\log K_{ow}$ indicates an increase in the potential bioconcentration in organisms (Net *et al.*, 2015).

Table 3: Plastic additives and their associated octanol-water partition coefficients (Log K_{ow}). Data were extracted from the following reviews: Bergé *et al.* (2012), Espinosa *et al.* (2016), Net *et al.* (2015) and Oehlmann *et al.* (2008)

Full name	Abbreviation	Log K _{ow}
butyl benzyl phthalate	BBP	4.70
di(2-ethylexyl) phthalate	DEHP	7.73
diethyl phthalate	DEP	2.54
diisobutyl phthalate	DiBP	4.27
diisodecyl phthalate	DiDP	9.46
diisononyl phthalate	DiNP	8.60
dimethyl phthalate	DMP	1.61
di- <i>n</i> -butyl phthalate	DnBP	4.27
di- <i>n</i> -octyl phthalate	DnOP	7.73
hexabromocyclododecane	HBCD	5.07 – 5.47
polybrominated diphenyl ether	PBDE	5.52 – 11.22
tetrabromobisphenol A	TBBPA	4.5
bisphenol A	BPA	3.40
nonylphenol	NP	4.48 – 4.80

2.1. Brominated flame retardants

Brominated flame retardants (BFR) are a class of additives used in plastic products to reduce flammability. These BFR are used in a variety of consumer products ranging from electronic devices to insulation foams. BFRs include a wide range of chemicals, of which polybrominated diphenyl ethers (PBDE), hexabromocyclododecane (HBCD – Pubchem ID: 18529) and tetrabromobisphenol A (TBBPA – Pubchem ID: 6618) (Talsness *et al.*, 2009) represent the main BFRs used in the plastic industry. These 3 classes (PBDE, HBCD and TBBPA) are described in details below. Lately, attention has been given to other emerging BFRs such as 1,2-bis(2,4,6-tribromophenoxy)ethane (BTBPE – Pubchem ID: 37840), decabromodiphenylethane (DBDPE – Pubchem ID: 10985889) and hexabromobenzene (HBB – Pubchem ID: 6905) as these have been identified in many environmental compartments, organisms, food and humans (European Food Safety Authority, 2012). As they are not chemically bound to the polymer matrix, they can leach into the surrounding environment

(Engler, 2012; Meeker *et al.*, 2009) with an exception for TBBPA which is chemically bound to the polymer (Morris *et al.*, 2004).

PBDEs are hydrophobic substances that include numerous formulations used in plastics as flame retardants. Indeed, there are three main commercial formulations called penta-, octa- and deca-BDEs (Chua *et al.*, 2014). These additives are ubiquitous, toxic, persistent and bioaccumulate in the environment and are of great concern for human health (Engler, 2012). As a result, penta- and octa-BDEs have been banned by the European Union since 2004 (European Directive, 2003), while deca-BDE was banned only in 2009 from electronic and electrical applications in the European Union (European Council Decision, 2009). These formulations can no longer be used in mixtures or products with a concentration higher than 0.1% by mass. In addition, tetra- to hepta-BDEs are listed for elimination in the Annex A of the Stockholm Convention on persistent organic pollutant (POP) (Stockholm Convention, 2016). Moreover, in Canada the use of tetra- to deca- BDE has been restricted under the SOR/2008-218 Regulations (Consolidated Regulation, 2008). Since 2006, penta- to octa- BDE have been subjected to a 90 day notification before importation or production in the US. Finally, deca-BDE importation and production have been entirely stopped (US Environmental Protection Agency, 2006, 2012) since 2013.

HBCD has three dominant stereoisomers: α -, β -, and γ -HBCD (European Food Safety Authority, 2011a). These BFRs are listed as POPs in the Stockholm Convention (Stockholm Convention, 2016) and the three isomers are subject to a request for authorization in the European Union (European Council Regulation, 2006). HBCDs are found in expanded PS (EPS) and extruded PS (XPS) up to 4-7% by weight (Al-Odaini *et al.*, 2015). Its use has been subjected to authorization in the European Union since 2006 in the annex XIV of the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) regulation (European Food Safety Authority, 2011a). Moreover, in 2013 HBCD was listed for elimination

in the Annex A of the Stockholm Convention with specific exemption for use and production in EPS and XPS (Cruz *et al.*, 2015; Stockholm Convention, 2016). In the US, Environmental Protection Agency conducted a risk assessment for HBCD according to the 2010 “Toxic Substances Control Act” action plan (US Environmental Protection Agency, 2010b).

TBBPA is produced by brominating bisphenol A (European Food Safety Authority, 2011b). According to the European Food Safety Authority (2011b), TBBPA is the most commonly produced BFR in the world and represents 60% of the BFR market. This BFR is used in acrylonitrile butadiene styrene (ABS) and in other plastic such as high impact PS and phenolic resin (Cruz *et al.*, 2015). Until now, no legislation concerning TBBPA has been applied in the European Union (Vandermeersch *et al.*, 2015).

2.2. Phthalates

Phthalic acid esters (PAE) or phthalates are a family of plastic additives used as plasticizers, mainly in PVC production (Arbeitsgemeinschaft PVC und Umwelt e.V, 2006). As a result, PVC can contain 10% to 60% phthalates by weight (Net *et al.*, 2015). As phthalates are not chemically bound to the polymer matrix, they can easily leach into the environment during manufacturing, use and disposal (Net *et al.*, 2015). PAEs have been found in a wide range of environments (as reviewed by Net *et al.* (2015)) and this is of concern, since some phthalates have been defined as endocrine disruptors, even at low concentrations (Oehlmann *et al.*, 2009).

In 2015, 8.4 million tons of plasticizers were used around the world, and di(2-ethylexyl) phthalate (DEHP – Pubchem ID: 8343) was the most commonly used plasticizer, representing 37.1% of the global plasticizer market (ECPI, 2016). Europe accounted for 1.3 million tons of the global plasticizer market in 2015 (ECPI, 2016), but DEHP was not the most commonly used plasticizer in Europe, as suggested by its 20% decrease in consumption between 1999 and 2004. DEHP has gradually been replaced by diisononyl phthalate (DiNP – Pubchem ID: 590836),

diisodecyl phthalate (DiDP – Pubchem ID: 33599) and di(2-Propyl Heptyl) phthalate (DPHP – Pubchem ID: 92344), which represented 57% of plasticizer consumption in Europe in 2015 ([Arbeitsgemeinschaft PVC und Umwelt e.V, 2006](#); [ECPI, 2016](#)).

2.3. Bisphenol A

BPA (Pubchem ID: 6623) is the most representative chemical of the bisphenol group and is one of the most commonly produced chemicals worldwide, with over three million tons produced annually ([Laing et al., 2016](#)). BPA is mainly used as a monomer for polycarbonate (PC) plastics (65% of volume used) and epoxy resins (30% of volume used), which are for instance the main component of the lining layer of aluminum cans ([Crain et al., 2007](#); [ICIS, 2003](#)). BPA can also be used as an antioxidant or as a plasticizer in other polymers (PP, PE and PVC) ([Rani et al., 2015](#)). Leaching of BPA can occur ([Sajiki and Yonekubo, 2003](#)), leading to the release of this additive from food and drink packaging, which is considered as a source of exposure for human beings ([Vandermeersch et al., 2015](#)). Despite its potential to leach from food packaging and the fact that it has been identified as a significant endocrine disruptor ([Oehlmann et al., 2009](#)), BPA is still allowed in the European Union for use in food contact material ([European Council Regulation, 2011](#)). Other bisphenol analogs, such as bisphenol B, bisphenol F and bisphenol S are used in plastics and may represent a threat to the environment even though their toxicity is still unknown ([Chen et al., 2016](#)).

2.4. Nonylphenols

Nonylphenols (NP) are intermediate products of the degradation of a widely used class of surfactants and antioxidants: nonylphenol ethoxylates (NPE) ([Engler, 2012](#)). NP and NPE are organic chemicals used for many applications such as paints, pesticides, detergents and personal care products (US Environmental Protection Agency, 2010a). They can also be used as antioxidants and plasticizers for the production of plastics ([Rani et al., 2015](#); US Environmental

Protection Agency, 2010a). Furthermore, NP have been found to leach out from plastic bottles to their water content (Loyo-Rosales *et al.*, 2004). Moreover, effluents from wastewater treatment plants are the major source of NP and NPE in the environment (Soares *et al.*, 2008). NP are considered as endocrine disruptors and their use is prohibited in the European Union due to their effects on the environment and human health (Rani *et al.*, 2015).

2.5. Antioxidants

Antioxidants are used as additives in many synthetic polymers including polyolefins (mainly PE and PP) which represent 60% of global demand for antioxidant additives (Höfer, 2012). Antioxidants are used to prevent the ageing of plastics and to delay oxidation (Lau and Wong, 2000). However, as with other plastic additives, antioxidants can leach out of the plastic and can migrate to food from plastic packaging and pose a threat in terms of food safety (Lau and Wong, 2000). Antioxidants from the Irganox® series are commonly used in plastics and they include Octadecyl 3-(3,5-di-*tert*-butyl-4-hydroxyphenyl)propionate (Irganox® 1076 – Pubchem: 16386), Pentaerythrityl-tetrakis-3-(3,5-di-*tert*-butyl-4-hydroxyphenyl)propionate (Irganox® 1010 – Pubchem ID: 64819) and 2,4-di-*tert*-butylphenol (Irgafos® 168 – Pubchem ID: 91601) (Lau and Wong, 2000).

Owing to the variety of plastic additives (BFR, phthalates, BPA, NP and antioxidants) used for plastic products conception and their detection in macro- and microplastic debris collected in environmental surveys, their occurrence in environmental matrices (water, sediment, biota) is expected and may pose major environmental concern as described below.

3. Plastic additives in the environment

3.1. Marine waters

Marine waters are affected by anthropogenic pressures as this natural compartment is the final receptacle of all discharge waters. Consequently, chemical pollutants including plastics additives have been detected in worldwide marine waters (Tables 4, 5 and 6) (Bergé *et al.*, 2013; Net *et al.*, 2015). Of all BFR, PBDE are the most commonly studied molecules in marine environments. PBDE have been widely found and multiple congeners have been monitored (Table 4). Concentrations varied from a few ng L⁻¹ to more than 10 ng L⁻¹ and congeners varied among the studied sites (Table 4).

Table 4: Concentrations of polybrominated diphenyl ether (PBDE) in seawater in ng L⁻¹

Location	ΣPBDE (ng L ⁻¹)	Range (ng L ⁻¹)	BDE congeners detected	Dominant congener	References
Port sea, Mediterranean Sea, Spain	23.2	4.2 – 19	BDE-28, -47	BDE-28	Sánchez-Avila <i>et al.</i> (2012)
Surface microlayer, China Sea, Hong-Kong	0.33	0.004 – 0.056	BDE-28, -47, -99, -100, -156, -183	BDE-156	Wurl <i>et al.</i> (2006)
Subsurface water, China Sea, Hong-Kong	0.1	0.002 – 0.082	BDE-28, -47, -99, -100, -183	BDE-47	Wurl <i>et al.</i> (2006)

Many studies on the contamination of the marine environment by phthalates showed concentrations ranging from a few pg L⁻¹ to around 10 µg L⁻¹, with DEHP being the most concentrated phthalate found in marine waters (Table 5).

237 **Table 5: Concentrations of phthalates in seawater in $\mu\text{g L}^{-1}$**

Location	DMP	DEP	DnBP	DiBP	BBP	DEHP	DnOP	Reference
Tees Bay, UK	$< 1 \times 10^{-3}$	0.025 – 0.5	0.47 – 0.55	0.66 – 1.1		0.98 – 2.2		Law et al. (1991)
North Sea, Germany	0.2×10^{-3}	0.67×10^{-2}	1.7×10^{-3}		0.05×10^{-3}	2.2×10^{-3}		Xie et al. (2005)
Surface waters, the Netherlands	0.004 – 0.49	0.07 – 2.3	$< 0.066 – 3.1$		0.001 – 1.8	0.9 - 5	0.002 – 0.078	Vethaak et al. (2005)
Bay of Biscay, Spain	$(7.5 \pm 0.4) \times 10^{-3}$	$(33 \pm 3) \times 10^{-3}$	$(83 \pm 7) \times 10^{-3}$		$(8 \pm 1) \times 10^{-3}$	$(64 \pm 4) \times 10^{-3}$	$(3.6 \pm 0.4) \times 10^{-3}$	Prieto et al. (2007)
Coastal seawater, Mediterranean Sea, Spain	0.003 – 0.14	0.024 – 0.48			0.001 – 0.10	0.03 – 0.62		Sánchez-Avila et al. (2012)
Port sea, Mediterranean Sea, Spain	0.004 – 0.012	0.024 – 0.87			0.003 – 0.80	0.06 – 5.97		Sánchez-Avila et al. (2012)
River – sea interface,	0.005	0.07 – 0.16			0.003 – 0.07	0.02 – 0.21		Sánchez-Avila et al. (2012)

Mediterranean Sea, Spain									
Ligurian Sea, Mediterranean Sea, Italy ¹						18.38 ± 44.39			Fossi <i>et al.</i> (2012)
Sardinian Sea, Mediterranean Sea, Italy ¹						23.42 ± 32.46			Fossi <i>et al.</i> (2012)
Barkley Sound, Canada			0.18 – 3.0			0.01 – 0.95			Keil <i>et al.</i> (2011)
Puget Sound, USA						0.06 – 0.64			Keil <i>et al.</i> (2011)
Klang River estuary, Australia						3.10 – 64.3			Tan (1995)
Caspian Sea, Iran	0.49	0.52							Hadjmohammadi <i>et al.</i> (2011)
Arctic	40 x 10 ⁻⁶	138 x 10 ⁻⁶	51 x 10 ⁻⁶	22 x 10 ⁻⁶	8 x 10 ⁻⁶	448 x 10 ⁻⁶			Xie <i>et al.</i> (2007)

Nonylphenol was detected in marine waters of Europe, Asia and North America (Bergé *et al.*, 2012; David *et al.*, 2009) (Table 6) and concentrations ranged from $0.2 \times 10^{-5} \mu\text{g L}^{-1}$ in the Sea of Japan to $4.6 \mu\text{g L}^{-1}$ in the Mediterranean Sea (Table 6). BPA, as for other additives, has been globally quantified in marine waters all around the world and concentrations ranged from ng L^{-1} in China to $\mu\text{g L}^{-1}$ in coastal waters of Singapore (Table 6).

Table 6: Ranges of concentrations of nonylphenol and BPA in seawater in $\mu\text{g L}^{-1}$

Chemicals	Location	Concentrations ($\mu\text{g L}^{-1}$)	References
Nonylphenol	German Bight, North Sea, Germany	0.006 – 0.033	Bester <i>et al.</i> (2001)
	Estuaries, the Netherlands	9×10^{-5} – 0.0014	Xie <i>et al.</i> (2006)
	Estuaries, UK	0.031 – 0.934	Jonkers <i>et al.</i> (2003)
	Mediterranean Sea, Spain	0.1 – 2.6	Blackburn <i>et al.</i> (1999)
	Venetian Lagoon, Italy	0.3 – 4.1	Petrovic <i>et al.</i> (2002)
	Jamaica Bay, US	0.004 – 0.211	Pojana <i>et al.</i> (2007)
	Masan Bay, South Korea	0.077 – 0.416	Ferguson <i>et al.</i> (2001)
	Sea of Japan, Japan	0.0097 – 0.928	Li <i>et al.</i> (2008)
		0.2×10^{-5} – 9.3×10^{-5}	Kannan <i>et al.</i> (1998)
	Surface waters, the Netherlands	0.009 - 1	Vethaak <i>et al.</i> (2005)
BPA		<0.012 – 0.33	Belfroid <i>et al.</i> (2002)
	Venetian Lagoon, Italy	<0.001 – 0.145	Pojana <i>et al.</i> (2007)
	Jiaozhou Bay, China	0.001 – 0.092	Fu <i>et al.</i> (2007)
	Estuaries, Japan	0.036 – 0.058	Kawahata <i>et al.</i> (2004)

Coastal waters, Singapore

0.01 – 2.47

Basheer *et al.*
(2004)

245

246 Overall, plastic additives have been detected worldwide in estuarine and marine waters at
247 concentrations ranging from pg/L to µg/L with PBDE and DEHP being the most commonly
248 reported congeners among BFR and phthalates, respectively. In addition to BPA and NP are
249 also frequently detected in seawater. As most of the plastic additives exhibit high K_{ow} , higher
250 concentrations are expected in sediment and marine organisms.

251 3.2. Sediment

252 As for marine waters, sediments are also affected by anthropogenic discharges and chemicals
253 including plastic additives. Regarding BFRs, multiple BDE congeners have been found in
254 marine sediments with BDE-209 being the major PBDE quantified in most studies at
255 concentrations ranging from ng/kg to mg/kg (Table 7). In the Netherlands, HBCD were also
256 found in sediments from the North Sea and Scheldt Estuary, respectively, at levels of 0.76 to
257 6.9 µg kg⁻¹ dry weight (dw) and 30 to 71 µg kg⁻¹ dw (Klamer *et al.*, 2005; Verslycke *et al.*,
258 2005).

259

260 **Table 7: Concentrations of polybrominated diphenyl ether (PBDE) in marine sediments in $\mu\text{g kg}^{-1}$ dry weight**

Location	ΣPBDE ($\mu\text{g kg}^{-1}$ dry weight)	Range ($\mu\text{g kg}^{-1}$ dry weight)	BDE congeners detected	Most abundant congener	References
North Sea, the Netherlands	126.3	0.4 - 32	BDE-28, -47, -66, -71, -75, -77, -85, -99, -100, -119, - 138, -153, -190, -209	BDE-209	Klamer <i>et al.</i> (2005)
Scheldt Estuary, the Netherlands	2198	0.2 – 1650	BDE-28, -47, -66, -85, -99, -100, -138, -153, -154, - 209	BDE-209	Verslycke <i>et al.</i> (2005)
Coastal waters, South Korea	27.8	0.0037 – 27.4	BDE-3, -7, -15, -28, -47, -49, -66, -71, -77, -85, -99, - 100, -119, -126, -138, -153, -154, -183, -209	BDE-209	Moon <i>et al.</i> (2007b)
Industrialized bays, South Korea	357.8	0.0012 - 283	BDE-3, -7, -15, -28, -47, -49, -66, -71, -77, -85, -99, - 100, -119, -126, -138, -153, -154, -183, -209	BDE-209	Moon <i>et al.</i> (2007a)

261

TBBPA was also found in the Scheldt Estuary at levels below $0.1 \mu\text{g kg}^{-1} \text{ dw}$ (Verslycke *et al.*, 2005). In their study, Klamer *et al.* (2005) also reported the presence of phthalates in North Sea sediments, namely dimethyl phthalate (DMP – Pubchem ID: 8554), diethyl phthalate (DEP – Pubchem ID: 6781), DBP, BBP, DEHP and DOP with DEHP being the most concentrated phthalate with 170 to $3,390 \mu\text{g kg}^{-1}$. Phthalates in marine sediments from the Gulf of Mexico were detected on average at 7.6 and $6.6 \mu\text{g kg}^{-1} \text{ dw}$ for di-*n*-butyl phthalate (DnBP – Pubchem ID: 3026) and DEHP respectively (Giam *et al.*, 1978). In Singapore Bay, phthalates reached 890 to $2,790 \mu\text{g kg}^{-1} \text{ dw}$ for DEHP (Chee *et al.*, 1996). For nonylphenol (Bergé *et al.*, 2012; David *et al.*, 2009), concentrations ranged from less than $1 \mu\text{g kg}^{-1} \text{ dw}$ in estuaries in the Netherlands to more than $20,000 \mu\text{g kg}^{-1} \text{ dw}$ in the sediments of Tokyo Bay (Table 8). Like BFRs, NP and phthalates, BPA has also been found worldwide in sediments (Table 8). Indeed, BPA concentrations ranged from a few $\mu\text{g kg}^{-1} \text{ dw}$ in Japan and China to hundreds of $\mu\text{g kg}^{-1} \text{ dw}$ in the Venetian Lagoon (Table 8).

Whether the plastic additives detected in marine sediments come from diffuse sources (wastewater, atmospheric deposition, sewage sludge, etc.) or leachate from plastic debris is unclear even though an increasing amount of evidence (Al-Odaini *et al.*, 2015) suggests that microplastic and plastic debris in general likely constitute sources of plastic additives in the marine environment.

281 **Table 8: Ranges of concentrations of nonylphenol and BPA in marine sediments in $\mu\text{g kg}^{-1}$ dry weight. nd: not detected.**

Localization	Nonylphenol range ($\mu\text{g kg}^{-1}$ dry weight)	BPA range ($\mu\text{g kg}^{-1}$ dry weight)	Reference
North Sea, Germany	13 – 55		Bester <i>et al.</i> (2001)
Estuaries, the Netherlands	0.9 – 1080		Jonkers <i>et al.</i> (2003)
Mediterranean Sea, Spain	18 – 590		Petrovic <i>et al.</i> (2002)
Venetian Lagoon, Italy	47 – 192		Pojana <i>et al.</i> (2007)
Jamaica Bay, US	7 – 13,700		Ferguson <i>et al.</i> (2001)
Southern California Bight, US	130 – 3200		Schlenk <i>et al.</i> (2005)
Masan Bay, South Korea	92 – 557		Li <i>et al.</i> (2008)
Tokyo Bay, Japan	142 – 20,700		Kurihara <i>et al.</i> (2007)
	120 – 640		Isobe <i>et al.</i> (2001)
The Netherlands		<1.1 – 43	Vethaak <i>et al.</i> (2005)
Venetian Lagoon, Italy		<2.0 – 118	Pojana <i>et al.</i> (2007)
Jiaozhou Bay, China		0.7 – 17	Fu <i>et al.</i> (2007)
Estuaries, Japan		nd – 2.7	Kawahata <i>et al.</i> (2004)

282 **3.3. Microplastics**

283 To date, only a few studies have focused on the detection of plastic additives from MP
284 collected in marine environments (Faure *et al.*, 2015; Fries *et al.*, 2013; Hirai *et al.*, 2011; Mato
285 *et al.*, 2001; Rani *et al.*, 2015; Rochman *et al.*, 2014). Mato *et al.* (2001) detected nonylphenols
286 in PP pellets deployed in Tokyo Bay and suggested that these compounds came from plastic
287 additives found in the PP pellets themselves. In another study, Hirai *et al.* (2011) measured high
288 concentrations of PBDEs, BPA and nonylphenols in PE and PP fragments collected on remote
289 or urban beaches and in the open ocean. It was stated that they originated from plastic additives
290 used for the manufacture of PP and PE. A wide range of plastic additives were also identified
291 using Pyrolysis-GC/MS with thermal desorption in MP collected from sediment of Norderney
292 Island (Fries *et al.*, 2013). MP particles were identified as PE, PP, PS and polyamide-6 (PA-6).

They were associated with DEHP, DnBP, diisobutyl phthalate (DiBP – Pubchem ID: 6782), and 2,4-di-*tert*-butylphenol (2,4-DTBP – Pubchem ID: 7311), used here as antioxidant additives for PE and PP, DnBP, DiBP, DEP and DMP for PS, and DEHP, and DiBP and DEP for PA-6 (Fries et al., 2013). Moreover, Rani et al. (2015) detected multiple plastic additives in plastic marine debris found on a beach in Geoje, South Korea. Indeed, the authors found BPA and phthalates in PP and PE plastic marine debris as well as antioxidants including Irganox 76 and 2,4-DTBP in PP and PE plastic marine debris. In a study dealing with plastic debris in the Atlantic Ocean, BPA, PBDEs and 4-nonylphenol were detected in plastic samples found at sea and the authors suggested that this chemical came mainly from plastic additives (Rochman et al., 2014). Moreover, some plastic additives were detected at concentrations up to 6 orders of magnitude higher than the concentrations measured in the surrounding water (Rochman et al., 2014). In a more recent study, Faure et al. (2015) quantified MP pollution in Swiss lakes and detected MP associated with plastic additives including PBDEs, NPs, BPA and phthalates at concentrations comparable to those reported in marine studies (from 10^{-1} to 10^6 ng g⁻¹) (Faure et al., 2015).

These six studies demonstrated that plastic additives, some of which are known to be potential endocrine disruptors, are quantifiable in MPs found in sediments or in marine waters, suggesting that leaching of additives occurs in the environment. This is clearly of great concern as microplastics exhibit a high propensity to enter all trophic levels due to their small size and ubiquity in marine environments, and given the fact that leaching may also occur in the digestive conditions of organisms upon MP ingestion.

4. Transfer and toxicity of plastics additives for marine organisms

4.1. Contamination of marine organisms by plastic additives

PBDE have been detected in tissues of numerous marine organisms such as bivalves ($\Sigma_{13}\text{BDE}$ ranged from 6.6 to 440 $\mu\text{g kg}^{-1}$ lipid weight) (Bellas *et al.*, 2014; Johansson *et al.*, 2006; Ramu *et al.*, 2007), fish ($\Sigma_7\text{BDE}$ ranged from 30.6 to 281 $\mu\text{g kg}^{-1}$ lipid weight) (Peng *et al.*, 2007) and mammals (around 100 $\mu\text{g kg}^{-1}$ wet weight (ww) in sperm whale (*Physeter microcephalus*) blubber) (de Boer *et al.*, 1998), suggesting that transfer from seawater, food or plastics to organisms occurs. In their work on the contamination of the Scheldt Estuary, Verslycke *et al.* (2005) found PBDE in sediment and in mysid shrimp (*Neomysis integer*) living in this estuary ($\Sigma_{10}\text{BDE}$: 2095 to 3562 ng g^{-1} lipid weight), and they highlighted that bioaccumulation was highest for BDE-47, -99 and -100 and lowest for BDE-209 because (i) highest brominated accumulate slower than lowest brominated congeners in the marine environment and (ii) they can be debrominated photolytically or biologically (Verslycke *et al.*, 2005). Phthalates (DMP, DEP, DiBP, DnBP, BBP, DEHP, DnOP, DnNP) were found in a wide range of organisms, including 18 different species ranging from primary producers (plankton and macroalgae) to picked dogfish (*Squalus acanthias*), but no biomagnification of the studied phthalates was observed through the food web (Mackintosh *et al.*, 2004). Recently, Cheng *et al.* (2013) also detected phthalates (DMP, DEP, dipropyl phthalate (DPRP – PubChem ID: 8559), DiBP, DnBP, 2-Methoxyethyl phthalate (DMEP – PubChem ID: 8344), DHP, BBP, DEHP, DOHP, DnOP, DNP+DiDP) in fish at concentrations ranging from 0.2 to 1.23 $\mu\text{g g}^{-1}$ ww (Cheng *et al.*, 2013). NP has been detected in many organisms commonly consumed as seafood products including oysters (*Crassostrea gigas*) (Cheng *et al.*, 2006), mussels (*Perna perna*) (Isobe *et al.*, 2007) and fishes (Ferrara *et al.*, 2008). For instance, Basheer *et al.* (2004) found NP and BPA in multiple fresh seafood products, including prawns (*Penaeus monodon*),

crabs (*Portunus pelagicus*), blood cockles (*Anadara granosa*), white clams (*Meretrix meretrix*), squid (*Loligo* sp.) and fish (*Decapterus russelli*), from a supermarket in Singapore.

Overall, these results suggest that contamination of marine organisms by plastic additives may occur *via* natural pathways (*i.e.* waterborne or foodborne exposure) or *via* ingestion of plastic debris including MP.

4.2. Plastics additive transfer to marine organisms

4.2.1. Evidence from laboratory experiments

To investigate the potential leaching of additives from MP in environments characterized by different conditions (pH, temperature, salinity, etc), several laboratory studies have been conducted over the last years. First, the influence of gut surfactant was tested on the desorption of adsorbed chemicals, including perfluorooctanoic acid (PFOA – Pubchem ID: 9554) and DEHP, from PVC and PE in a study undertaken by [Bakir *et al.* \(2014\)](#). Desorption was higher in gut surfactant at 38°C (*i.e.* warm blooded animals) than in gut surfactant at 18°C (*i.e.* cold blooded animals) and in seawater at 18°C for DEHP. PFOA exhibited low affinity for PVC or PE regardless of the surfactant ([Bakir *et al.*, 2014](#)). The same authors suggested that the passage of plastic through the gut could enhance desorption of pollutants and act as a transfer route for accumulation of these pollutants. However, in a more recent study, [Bakir *et al.* \(2016\)](#) demonstrated, using a one-compartment model, that MP do not provide an additional pathway for the transfer of adsorbed chemicals, including DEHP and PFOA, from seawater to marine organisms even if MP transits through the gut. Some laboratory studies have used MP in the presence of additives to determine if these chemicals can transfer to organisms. For instance, [Chua *et al.* \(2014\)](#) exposed the marine amphipod *Allorchestes compressa* to PBDE in the presence or absence of microbeads with PBDEs adsorbed on microbeads. Both microbead ingestion and PBDE transfer *via* the microbeads were demonstrated at the end of the exposure.

However, concentrations of PBDEs were lower in amphipods exposed to PBDE adsorbed on microbeads than in amphipods exposed to PBDEs without microbeads (Chua *et al.*, 2014) suggesting that transfer of PBDE adsorbed on MP can occur but at a lesser extent than the transfer via water. Similarly, Wardrop *et al.* (2016) exposed rainbow fish (*Melanotaenia fluviatilis*) to microbeads spiked with PBDEs (BDE-28, -47, -100, -99, -153, -154, -183 and -209) and compared them to control fish and fish exposed to microbeads alone. Here, PBDEs were analyzed in fish tissues excluding the stomach, liver, gall bladder and gonads to exclude spiked microbeads from the PBDEs analyses. During exposure, fish exposed to microbeads spiked with PBDEs showed higher concentrations than the two controls, and lower brominated congeners were better transferred in fish tissues than higher brominated congeners. On the other hand control fish and fish exposed to PBDE-free microbeads showed the same low levels of PBDEs concentration in their tissues suggesting that MP do not reduce contaminant body burden as it was previously hypothesized (Koelmans *et al.*, 2013a, b; Koelmans *et al.*, 2016). More realistic experiments have been performed using plastics incubated or collected in natural environments. For instance Rochman *et al.* (2013) used low-density polyethylene (LDPE) pellets deployed in seawater for two months and showed that the LDPE pellets adsorbed chemicals from the surrounding environment. Exposure of Japanese medaka (*Oryzias latipes*) to these pellets resulted in the accumulation of significant amounts of PBDEs and was associated with liver toxicity and pathology including glycogen depletion and cell necrosis for example (Rochman *et al.*, 2013). Bioaccumulation of PBDEs was also demonstrated in a terrestrial invertebrate, the house cricket (*Acheta domesticus*), as a result of PUR foam ingestion (Gaylor *et al.*, 2012). Another laboratory study demonstrated that the transfer of nonylphenol, triclosan and PBDE-47 can occur via MP ingestion in the lugworm (*Arenicola marina*) with possible effects on lugworm behavior (Browne *et al.*, 2013).

Overall, these laboratory experiments demonstrated transfer of plastic additives upon MP ingestion, sometimes in association with toxicity or behavior change.

4.2.2. Evidence from field studies

Levels of accumulated plastic additives in the environment or organisms have often been considered as a proxy indicator of plastic exposure in the marine environment as a consequence of the release of additives from plastic debris. For instance, a study on *Puffinus tenuirostris* showed that this bird ingested plastics at sea and that these plastics transferred flame retardant additives (PBDE) including BDE-209, which is specific to plastic (Tanaka *et al.*, 2013). In another study, the authors demonstrated that the transfer of PBDE may occur mainly by plastic ingestion through exposure by prey ingestion (Tanaka *et al.*, 2015). In another study, Rochman *et al.* (2014) examined the possible relationship between plastic densities at sea and levels of chemicals in fish inhabiting those areas. The results showed that PBDEs, especially the highest brominated congeners (BDE-209), may be an indicator of plastic pollution as previously suggested (Tanaka *et al.*, 2013). In the North Pacific Gyre, yellowtail (*Seriola lalandi*) were sampled to evaluate levels of plastic in the stomach and concentrations of pollutants and additives in their tissues (Gassel *et al.*, 2013). Ten percent of the yellowtail had ingested plastics, and PBDE and nonylphenol were concomitantly found in the fish tissues (Gassel *et al.*, 2013). Gassel *et al.* (2013) suggested that contamination of fish by nonylphenol and PBDE-209 could originate from the ingested plastic as mentioned above (Hirai *et al.*, 2011; Rochman *et al.*, 2014; Tanaka *et al.*, 2013; Teuten *et al.*, 2009). Other chemicals are also used as proxies for MP contamination such as DEHP (Fossi *et al.*, 2012; Fossi *et al.*, 2014; Fossi *et al.*, 2016). More recently, a study showed a higher accumulation of HBCDs in mussels (*Mytilus galloprovincialis*) inhabiting styrofoam debris (EPS) in comparison with mussels living on other plastic debris or rocks (Jang *et al.*, 2016). The authors also suggested that the isomeric

profiles of detected HBCDs support the transfer of this flame retardant from the styrofoam debris to mussels through ingestion of EPS particles.

Field surveys showed that MP ingestion may constitute another route of transfer of plastic additives in marine organisms, leading to the use of plastic additives tissue content (mainly BDE-209 and DEHP) as a proxy for plastic exposure or ingestion.

4.3. Toxicity of plastic additives demonstrated by leaching experiments

Evidence for plastic toxicity has been rising in the last years. While direct toxicity can occur due to the physical impacts of plastic ingestion (for a review, see [Wright *et al.* \(2013\)](#)), indirect toxicity may be observed in relation to the release of hazardous chemicals from plastics. As most plastic additives are not chemically but physically bound to the plastic, they can be released into the environment and become available to organisms. Recent studies demonstrated, using leaching experiments, that various plastics are toxic to a wide range of organisms (Table 9).

424 **Table 9: list of aquatic species, plastic polymer types, exposure times and endpoints used in various leaching experiments**

Species	Plastic type	Exposure time	Exposure level	Endpoints	Reference
<i>Daphnia magna</i>	PC, PVC, PU, PE, LDPE, PMMA, PET, HDPE, PTFE, ABS, PP, MDPE	24 and 48 hours	70 – 100 g L ⁻¹	Mortality	Lithner et al. (2009)
<i>Daphnia magna</i>	PP, HDPE, PVC, ABS, Epoxy resin	24 and 48 hours	Up to 250 g L ⁻¹	Mortality	Lithner et al. (2012)
<i>Nitroca sinipes</i>	PP, PVC, PS, PET, PUR, LDPE, HDPE, ABS, PLA, Unknown	96 hours	100 g L ⁻¹	Mortality	Bejgarn et al. (2015)
<i>Amphibalanus amphitrite</i>	PET, HDPE, PVC, LDPE, PP, PS, PC	24, 48, 72 and 96 hours	0.1 – 0.5 m ² L ⁻¹	Settlement	Li et al. (2016)
<i>Perna perna</i>	Virgin (PP) and beached pellets	48 hours	25% of pellets (v/v)	Embryo development	Gandara e Silva et al. (2016)
<i>Pseudochromis fridmani</i>	PE (two different origins)	48 hours	-	Mortality	Hamlin et al. (2015)
ABS: Acrylonitrile butadiene styrene; PC: Polycarbonate; PE: Polyethylene; LDPE: Low-Density Polyethylene; MDPE: Medium-Density Polyethylene; HDPE: High-Density Polyethylene; PET: Polyethylene terephthalate; PLA: Poly Lactic Acid; PMMA: Polymethyl Methacrylate; PP: Polypropylene; PTFE: Polytetrafluoroethylene; PU: Polyurethane; PVC: Polyvinyl Chloride					

425

Li *et al.* (2016) used the seven categories of recyclable plastics (High Density PE (HDPE), LDPE, PP, PVC, Polycarbonate (PC), PET and PS) to quantify the impact of their leachate on the survival and settlement of barnacle *Amphibalanus amphitrite* larvae. Leachates were prepared by placing 1 x 1 cm pieces of each plastic in 20 mL of filtered seawater for 24h at 28°C (Li *et al.*, 2016). Survival was significantly lowered at the highest leachate concentration (0.10 and 0.50 m² of plastic material L⁻¹) for all plastics and PVC was the most toxic plastic for *A. Amphitrite* larvae. Additionally, settlement was also inhibited with all plastics leachates (Li *et al.*, 2016). Similarly, Bejgarn *et al.* (2015) exposed the copepod *Nitocra sinipes* to the leachate of weathered or non-weathered plastics. Here, leaching experiments were performed with leachates prepared with 10 g of each plastic placed in 100 mL of brackish seawater from the Baltic rotating at 6-21 rpm for 72h in the dark (Bejgarn *et al.*, 2015). Of the twenty-one plastics tested, eight (DVD-case (PP), biodegradable bag, costume- (PVC), flyswatter packaging (PVC), computer housing (unknown), garden hose (PVC), car dashboard (unknown) and phone cover (PUR)) demonstrated toxicity (mortality after 96h) to *N. sinipes*, and after weathering, toxicity either increased or decreased depending on the plastics (Bejgarn *et al.*, 2015). Two leaching studies were carried out on the copepod *Daphnia magna* (Lithner *et al.*, 2012; Lithner *et al.*, 2009), a common organism used in ecotoxicological studies. Lithner *et al.* (2009) prepared their leachates by placing plastic pieces in deionized water to obtain a liquid to solid ratio of 10 L kg⁻¹ which was horizontally shaken at 60 rpm for 24h (16h of fluorescent light and 8h of darkness) at 20°C. Out of the thirty-two plastics tested only nine, including five composed of PVC, showed acute toxicity (immobility after 24 and 48h ; EC₅₀ ranging from 5 to 80 g plastic material L⁻¹) to *D. magna* and it has been suggested that the toxicity of PVC was due to the phthalate content (Lithner *et al.*, 2009). In a second study, Lithner *et al.* (2012) used PP, PE, PVC, acrylonitrile-butadiene-styrene (ABS) and epoxy resin, and they prepared their leachates by adding 250 g of plastic in 1 L of deionized water shaken at 90 rpm at 50°C for 3

days. As previously demonstrated, PVC caused acute toxicity (immobility after 24 and 48h ; EC₅₀ ranging from 2-235 g plastic material L⁻¹) probably in relation to its phthalate leachates, however the acute toxicity observed with the epoxy resin was not attributed to a specific chemical compound (Lithner *et al.*, 2012). A more recent study evaluated the toxicity of virgin and beached pellets on the embryo development of brown mussels (*Perna perna*) (Gandara e Silva *et al.*, 2016). Here, the authors exposed the brown mussel embryo to 2 mL of virgin (PP) or beached (42% PE and 58% unknown composition) pellets in 8 mL of seawater, and toxicity was assessed by determining the percentage of dead or abnormal embryos (Gandara e Silva *et al.*, 2016). The leaching experiment led to 23.5% and 100% dead or abnormal embryos for virgin and beached pellets, respectively. It has been suggested that the difference in toxicity was mainly due to the difference in the chemical load of the pellets used (Gandara e Silva *et al.*, 2016). Beached pellets were exposed to *in situ* contamination leading to adsorption of pollutants and to additives already found inside the polymeric matrix. These leaching experiments showed that plastic leachates and especially PVC leachates (*i.e.* phthalates) can lead to adverse effects on organisms. However, the toxicity highlighted in these five experiments was not attributed to specific chemical compounds (Bejgarn *et al.*, 2015; Gandara e Silva *et al.*, 2016; Li *et al.*, 2016; Lithner *et al.*, 2012; Lithner *et al.*, 2009). As suggested by Li *et al.* (2016), chemical identification should be undertaken during leaching experiments with a focus on plastic additives in order to identify the compound or its degradation products responsible for the observed toxicity. For instance, a more recent leaching study focused on the effects of nonylphenol on the coral reef fish *Pseudochromis fridmani* by exposing single fish to the leachate of plastic bags made of two PE (PE1 and PE2) from different manufacturers for 48h (Hamlin *et al.*, 2015). Nonylphenol leached in the water at higher concentrations for PE2 than for PE1; with respectively 41.0 ± 5.5 and 2.5 ± 0.2 $\mu\text{g L}^{-1}$, and 60% and 11% of the fish died after 48 hours of exposure to leachates from PE2 and PE1, respectively. However, Hamlin

et al. (2015) only focused their work on nonylphenol and did not study PE1 and PE2 compositions in terms of other additive contents. This study demonstrated that exposing fish to two identical plastic polymers (PE) may result in drastically different outcomes, as the plastic additives incorporated in each plastics are dependent on the plastic manufacturer and most of the time, their exact compositions remain unknown (Hamlin *et al.*, 2015). Studies are required to explore potential differences between plastics from different manufacturers and toxicity related to the diversity of chemicals used in the plastic industry.

Exposure experiments based on leaching processes conducted on a wide range of polymers and target organisms confirmed toxicity of plastics additives, which highlights the need for non-target screening analysis covering a broad range of chemicals in order to better identify the main compound(s) responsible for the toxicity.

4.4. Relative importance of HOC in comparison with plastic additives: case of modelling studies

The high affinity of plastic polymers for HOC has been demonstrated in numerous laboratory experiments (Bakir *et al.*, 2014; Teuten *et al.*, 2007), and an increasing number of studies have focused on the role of MP as a vector for HOC in marine organisms (Besseling *et al.*, 2013; Rochman *et al.*, 2013). However, recent studies have suggested that given (i) baseline contamination levels of seawater and marine organisms, and (ii) the low abundance of plastic particles relative to other suspended particles found in oceans (such as organic matter, plankton, detritus etc), exposure to HOC *via* plastic may be negligible compared to natural pathways (Bakir *et al.*, 2016; Beckingham and Ghosh, 2017; Koelmans *et al.*, 2013a, b; Koelmans *et al.*, 2016; Paul-Pont *et al.*, 2016). Moreover, Koelmans *et al.* (2016) suggested that MPs ingestion by marine biota does not increase their exposure to HOCs and could have a cleaning effect while concerns have arisen regarding risk due to plastic additives.

So far most modelling studies have focused their work on adsorbed HOC (Bakir *et al.*, 2016; Koelmans *et al.*, 2016). However, no model is yet available on the transport and fate of plastic additives leaching from plastic debris although (i) plastic additives can be added in very high concentration depending on the application; and (ii) transfer of plastic additives to marine organism upon plastic ingestion has been demonstrated both in laboratory experiments and in field studies. It highlights the need to include these chemicals in future modelling work in order to better clarify their potential for transfer.

5. Conclusion

Plastic additives associated with MP have received less attention than HOC adsorbed on MPs and the present review highlighted the need for upcoming studies to better characterize plastic additives associated with microplastics found at sea as well as their potential release in environmental matrices. As PE and PP are the main plastic debris found at sea, these two polymers should be investigated alongside with PVC due to its particularly high concentration in hazardous additives. Non-target screening analysis is required to identify the broad range of plastic additives leaching from these polymers and to better identify the main compound(s) responsible for toxicity. In addition, special attention should definitely be paid to hazardous plastic additives known to be major endocrine disruptor, namely bisphenol A and phthalates. Experimental and modelling studies are required to better characterize (i) the transfer of plastic additives upon MP ingestion relative to waterborne and foodborne exposure, and (ii) the effects of plastic additives on marine organisms. Such experiments should be realized using standardized “laboratory-made” MP, in which plastic additives are well characterized and introduced in controlled amounts reflecting industrial processes. The impacts of ageing plastic under realistic conditions on the transfer of plastic additives also need to be evaluated to investigate more environmentally relevant scenarios.

Acknowledgments

Ludovic Hermabessiere is grateful to the Hauts-de-France Region and ANSES (French Agency for Food, Environmental and Occupational Health & Safety) for the financial support of his PhD. This work was funded by ANSES through a “*Convention de Recherche et Développement*” (2015-CRD-11). This paper was also supported by the CESA of the French National Research Agency (ANR) (ANR-15-CE34-0006-02), as part of the NANOPLASTICS project. Authors also thank James Marchant for his helpful revision of English and his comments on the manuscript.

References

- Al-Odaini, N.A., Shim, W.J., Han, G.M., Jang, M., Hong, S.H., 2015. Enrichment of hexabromocyclododecanes in coastal sediments near aquaculture areas and a wastewater treatment plant in a semi-enclosed bay in South Korea. *Science of The Total Environment*. 505, 290-298. doi: 10.1016/j.scitotenv.2014.10.019
- Andrady, A.L., 2011. Microplastics in the marine environment. *Marine Pollution Bulletin*. 62, 1596-1605. doi: 10.1016/j.marpolbul.2011.05.030
- Andrady, A.L., Neal, M.A., 2009. Applications and societal benefits of plastics. *Philosophical Transactions of the Royal Society B: Biological Sciences*. 364, 1977-1984. doi: 10.1098/rstb.2008.0304
- Arbeitsgemeinschaft PVC und Umwelt e.V., 2006. Plasticizers market data. Available on: http://www.pvc-partner.com/fileadmin/user_upload/downloads/Weichmacher/Marktdaten_Weichmacher_230106.lin_en.pdf, Accessed on: 05/31/2016
- Arthur, C., Baker, J., Bamford, H., 2009. International Research Workshop on the Occurrence, Effects, and Fate of Microplastic Marine Debris. NOAA Technical Memorandum NOS-OR&R-30.
- Bakir, A., Rowland, S.J., Thompson, R.C., 2014. Enhanced desorption of persistent organic pollutants from microplastics under simulated physiological conditions. *Environmental Pollution*. 185, 16-23. doi: 10.1016/j.envpol.2013.10.007
- Bakir, A., O'Connor, I.A., Rowland, S.J., Hendriks, A.J., Thompson, R.C., 2016. Relative importance of microplastics as a pathway for the transfer of hydrophobic organic chemicals to marine life. *Environmental Pollution*. 219, 56-65. doi: 10.1016/j.envpol.2016.09.046
- Barnes, D.K., Galgani, F., Thompson, R.C., Barlaz, M., 2009. Accumulation and fragmentation of plastic debris in global environments. *Philosophical Transactions of the Royal Society B: Biological Sciences*. 364, 1985-1998. doi: 10.1098/rstb.2008.0205
- Basheer, C., Lee, H.K., Tan, K.S., 2004. Endocrine disrupting alkylphenols and bisphenol-A in coastal waters and supermarket seafood from Singapore. *Marine pollution bulletin*. 48, 1161-1167. doi: 10.1016/j.marpolbul.2004.04.009
- Beckingham, B., Ghosh, U., 2017. Differential bioavailability of polychlorinated biphenyls associated with environmental particles: Microplastic in comparison to wood, coal and biochar. *Environmental Pollution*. 220, 150-158. doi: 10.1016/j.envpol.2016.09.033
- Bejgarn, S., MacLeod, M., Bogdal, C., Breitholtz, M., 2015. Toxicity of leachate from weathering plastics: An exploratory screening study with *Nitocra spinipes*. *Chemosphere*. 132, 114-119. doi: 10.1016/j.chemosphere.2015.03.010
- Belfroid, A., van Velzen, M., van der Horst, B., Vethaak, D., 2002. Occurrence of bisphenol A in surface water and uptake in fish: evaluation of field measurements. *Chemosphere*. 49, 97-103. doi: 10.1016/S0045-6535(02)00157-1
- Bellas, J., Albentosa, M., Vidal-Liñán, L., Besada, V., Franco, M.Á., Fumega, J., González-Quijano, A., Viñas, L., Beiras, R., 2014. Combined use of chemical, biochemical and physiological variables in mussels for the assessment of marine pollution along the N-NW Spanish coast. *Marine Environmental Research*. 96, 105-117. doi: 10.1016/j.marenvres.2013.09.015
- Bergé, A., Cladière, M., Gasperi, J., Coursimault, A., Tassin, B., Moilleron, R., 2012. Meta-analysis of environmental contamination by alkylphenols. *Environmental Science and Pollution Research*. 19, 3798-3819. doi: 10.1007/s11356-012-1094-7

- 580 Bergé, A., Cladière, M., Gasperi, J., Coursimault, A., Tassin, B., Moilleron, R., 2013. Meta-
581 analysis of environmental contamination by phthalates. *Environmental Science and*
582 *Pollution Research*. 20, 8057-8076. doi: 10.1007/s11356-013-1982-5
- 583 Besseling, E., Wegner, A., Foekema, E.M., van den Heuvel-Greve, M.J., Koelmans, A.A.,
584 2013. Effects of Microplastic on Fitness and PCB Bioaccumulation by the Lugworm
585 *Arenicola marina* (L.). *Environmental science & technology*. 47, 593-600. doi:
586 10.1021/es302763x
- 587 Bester, K., Theobald, N., Schröder, H.F., 2001. Nonylphenols, nonylphenol-ethoxylates, linear
588 alkylbenzenesulfonates (LAS) and bis (4-chlorophenyl)-sulfone in the German Bight of
589 the North Sea. *Chemosphere*. 45, 817-826. doi: 10.1016/S0045-6535(01)00023-6
- 590 Blackburn, M.A., Kirby, S.J., Waldock, M.J., 1999. Concentrations of alkylphenol
591 polyethoxylates entering UK estuaries. *Marine pollution bulletin*. 38, 109-118. doi:
592 10.1016/S0025-326X(98)00104-0
- 593 Boerger, C.M., Lattin, G.L., Moore, S.L., Moore, C.J., 2010. Plastic ingestion by planktivorous
594 fishes in the North Pacific Central Gyre. *Marine pollution bulletin*. 60, 2275-2278. doi:
595 10.1016/j.marpolbul.2010.08.007
- 596 Bouwmeester, H., Hollman, P.C., Peters, R.J., 2015. Potential Health Impact of
597 Environmentally Released Micro-and Nanoplastics in the Human Food Production
598 Chain: Experiences from Nanotoxicology. *Environmental science & technology*. 49,
599 8932-8947. doi: 10.1021/acs.est.5b01090
- 600 Browne, Mark A., Niven, Stewart J., Galloway, Tamara S., Rowland, Steve J., Thompson,
601 Richard C., 2013. Microplastic Moves Pollutants and Additives to Worms, Reducing
602 Functions Linked to Health and Biodiversity. *Current Biology*. 23, 2388-2392. doi:
603 10.1016/j.cub.2013.10.012
- 604 Browne, M.A., Crump, P., Niven, S.J., Teuten, E., Tonkin, A., Galloway, T., Thompson, R.,
605 2011. Accumulation of microplastic on shorelines worldwide: sources and sinks.
606 *Environmental science & technology*. 45, 9175-9179. doi: 10.1021/es201811s
- 607 Carpenter, E.J., Anderson, S.J., Harvey, G.R., Miklas, H.P., Peck, B.B., 1972. Polystyrene
608 Spherules in Coastal Waters. *Science*. 178, 749-750. doi: 10.1126/science.178.4062.749
- 609 Carr, S.A., Liu, J., Tesoro, A.G., 2016. Transport and fate of microplastic particles in
610 wastewater treatment plants. *Water Research*. 91, 174-182. doi:
611 10.1016/j.watres.2016.01.002
- 612 Chang, M., 2015. Reducing microplastics from facial exfoliating cleansers in wastewater
613 through treatment versus consumer product decisions. *Marine pollution bulletin*. 101,
614 330-333. doi: 10.1016/j.marpolbul.2015.10.074
- 615 Chee, K.K., Wong, M.K., Lee, H.K., 1996. Microwave extraction of phthalate esters from
616 marine sediment and soil. *Chromatographia*. 42, 378-384. doi: 10.1007/bf02272126
- 617 Chen, D., Kannan, K., Tan, H., Zheng, Z., Feng, Y.-L., Wu, Y., Widelka, M., 2016. Bisphenol
618 Analogues Other Than BPA: Environmental Occurrence, Human Exposure, and
619 Toxicity—A Review. *Environmental science & technology*. 50, 5438-5453. doi:
620 10.1021/acs.est.5b05387
- 621 Cheng, C.-Y., Liu, L.-L., Ding, W.-H., 2006. Occurrence and seasonal variation of alkylphenols
622 in marine organisms from the coast of Taiwan. *Chemosphere*. 65, 2152-2159. doi:
623 10.1016/j.chemosphere.2006.06.017
- 624 Cheng, Z., Nie, X.-P., Wang, H.-S., Wong, M.-H., 2013. Risk assessments of human exposure
625 to bioaccessible phthalate esters through market fish consumption. *Environment*
626 *International*. 57–58, 75-80. doi: 10.1016/j.envint.2013.04.005
- 627 Chua, E.M., Shimeta, J., Nugegoda, D., Morrison, P.D., Clarke, B.O., 2014. Assimilation of
628 Polybrominated Diphenyl Ethers from Microplastics by the Marine Amphipod,

- 629 *Allorchestes Compressa*. Environmental science & technology. 48, 8127-8134. doi:
630 10.1021/es405717z
- 631 Cole, M., Lindeque, P., Halsband, C., Galloway, T.S., 2011. Microplastics as contaminants in
632 the marine environment: a review. Marine pollution bulletin. 62, 2588-2597. doi:
633 10.1016/j.marpolbul.2011.09.025
- 634 Consolidated Regulation, 2008. Polybrominated Diphenyl Ethers Regulations (SOR/2008-
635 218). Available on: <http://laws-lois.justice.gc.ca/eng/regulations/SOR-2008-218/>,
636 Accessed on: 04/26/2017
- 637 Cózar, A., Echevarría, F., González-Gordillo, J.I., Irigoien, X., Úbeda, B., Hernández-León, S.,
638 Palma, Á.T., Navarro, S., García-de-Lomas, J., Ruiz, A., Fernández-de-Puelles, M.L.,
639 Duarte, C.M., 2014. Plastic debris in the open ocean. Proceedings of the National
640 Academy of Sciences. 111, 10239-10244. doi: 10.1073/pnas.1314705111
- 641 Crain, D.A., Eriksen, M., Iguchi, T., Jobling, S., Laufer, H., LeBlanc, G.A., Guillette Jr, L.J.,
642 2007. An ecological assessment of bisphenol-A: Evidence from comparative biology.
643 Reproductive Toxicology. 24, 225-239. doi: 10.1016/j.reprotox.2007.05.008
- 644 Cruz, R., Cunha, S.C., Casal, S., 2015. Brominated flame retardants and seafood safety: A
645 review. Environment International. 77, 116-131. doi: 10.1016/j.envint.2015.01.001
- 646 Dantas, D.V., Barletta, M., da Costa, M.F., 2012. The seasonal and spatial patterns of ingestion
647 of polyfilament nylon fragments by estuarine drums (*Sciaenidae*). Environmental
648 Science and Pollution Research. 19, 600-606. doi: 10.1007/s11356-011-0579-0
- 649 David, A., Fenet, H., Gomez, E., 2009. Alkylphenols in marine environments: Distribution
650 monitoring strategies and detection considerations. Marine pollution bulletin. 58, 953-
651 960. doi: 10.1016/j.marpolbul.2009.04.021
- 652 de Boer, J., Wester, P.G., Klammer, H.J.C., Lewis, W.E., Boon, J.P., 1998. Do flame retardants
653 threaten ocean life? Nature. 394, 28-29.
- 654 de los Ríos, A., Juanes, J.A., Ortiz-Zarragoitia, M., López de Alda, M., Barceló, D., Cajaraville,
655 M.P., 2012. Assessment of the effects of a marine urban outfall discharge on caged
656 mussels using chemical and biomarker analysis. Marine pollution bulletin. 64, 563-573.
657 doi: 10.1016/j.marpolbul.2011.12.018
- 658 De Witte, B., Devriese, L., Bekaert, K., Hoffman, S., Vandermeersch, G., Cooreman, K.,
659 Robbens, J., 2014. Quality assessment of the blue mussel (*Mytilus edulis*): Comparison
660 between commercial and wild types. Marine pollution bulletin. 85, 146-155. doi:
661 10.1016/j.marpolbul.2014.06.006
- 662 Devriese, L.I., van der Meulen, M.D., Maes, T., Bekaert, K., Paul-Pont, I., Frère, L., Robbens,
663 J., Vethaak, A.D., 2015. Microplastic contamination in brown shrimp (*Crangon*
664 *crangon*, Linnaeus 1758) from coastal waters of the Southern North Sea and Channel
665 area. Marine pollution bulletin. 98, 179-187. doi: 10.1016/j.marpolbul.2015.06.051
- 666 Dris, R., Gasperi, J., Saad, M., Mirande, C., Tassin, B., 2016. Synthetic fibers in atmospheric
667 fallout: A source of microplastics in the environment? Marine pollution bulletin. 104,
668 290-293. doi: 10.1016/j.marpolbul.2016.01.006
- 669 ECPI, 2016. Plasticisers. Available on: http://www.plasticisers.org/en_GB/plasticisers,
670 Accessed on: 11/24/2016
- 671 Engler, R.E., 2012. The Complex Interaction between Marine Debris and Toxic Chemicals in
672 the Ocean. Environmental science & technology. 46, 12302-12315. doi:
673 10.1021/es3027105
- 674 Eriksen, M., Lebreton, L.C., Carson, H.S., Thiel, M., Moore, C.J., Borroero, J.C., Galgani, F.,
675 Ryan, P.G., Reisser, J., 2014. Plastic pollution in the world's oceans: more than 5 trillion
676 plastic pieces weighing over 250,000 tons afloat at sea. PloS one. 9, e111913.

- Eriksson, C., Burton, H., 2003. Origins and Biological Accumulation of Small Plastic Particles in Fur Seals from Macquarie Island. *AMBIO: A Journal of the Human Environment*. 32, 380-384. doi: 10.1579/0044-7447-32.6.380
- Espinosa, C., Esteban, M.Á., Cuesta, A., 2016. Microplastics in Aquatic Environments and Their Toxicological Implications for Fish in Soloneski, S., Larramendy, M.L. (Eds.), *Toxicology - New Aspects to This Scientific Conundrum*. InTech, pp. 220.
- European Chemical Agency, 2017. REACH. Available on: <https://echa.europa.eu/regulations/reach>, Accessed on: 04/24/2017
- European Council Decision, 2009. Official Journal of the European Union Commission. Decision 2005/717/EC-Exemption of Deca BDE from the prohibition on Use, C116. May 9, 2008. Available on: <http://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A62006CJ0014>, Accessed on: 09/12/2016
- European Council Regulation, 2006. European Commission Regulation No.1907/2006 Concerning the Registration Evaluation Authorisation and Restriction of Chemicals (REACH) establishing a European Chemicals Agency Amending Directive 1999/45/EC and Repealing Council Regulation (EEC) No.793/93 and Commission Regulation (EC) No.1488/94 as well as Council Directive 76/769/EEC and Commission Directives 91/155/EEC93/67/EEC93/105/EC and 2000/21/EC. December 18,2006. Available on: <http://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:02006R1907-20140410&from=FR>, Accessed on: 09/12/2016
- European Council Regulation, 2011. European Commission Regulation No.10/2011 on Plastic Materials and Articles Intended to Come into Contact with Food. Available on: <http://eur-lex.europa.eu/legal-content/EN/ALL/?uri=CELEX%3A32011R0010>, Accessed on: 09/14/2016
- European Directive, 2003. Directive 2003/11/EC of the European Parliament and of the Council of 6 February 2003 amending for the 24th time Council Directive 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations (pentabromodiphenyl ether, octabromo-diphenyl ether). Available on: <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2003:042:0045:0046:EN:PDF>, Accessed on: 09/12/2016
- European Food Safety Authority, 2011a. Scientific Opinion on Hexabromocyclododecanes (HBCDDs) in Food. *EFSA Journal*. 9, n/a-n/a. doi: 10.2903/j.efsa.2011.2296
- European Food Safety Authority, 2011b. Scientific Opinion on Tetrabromobisphenol A (TBBPA) and its derivatives in food. *EFSA Journal*. 9, n/a-n/a. doi: 10.2903/j.efsa.2011.2477
- European Food Safety Authority, 2012. Scientific Opinion on Emerging and Novel Brominated Flame Retardants (BFRs) in Food. *EFSA Journal*. 10, n/a-n/a. doi: 10.2903/j.efsa.2012.2908
- Faure, F., Demars, C., Wieser, O., Kunz, M., de Alencastro, L.F., 2015. Plastic pollution in Swiss surface waters: nature and concentrations, interaction with pollutants. *Environmental Chemistry*. 12, 582-591. doi: 10.1071/EN14218
- Fendall, L.S., Sewell, M.A., 2009. Contributing to marine pollution by washing your face: Microplastics in facial cleansers. *Marine pollution bulletin*. 58, 1225-1228. doi: 10.1016/j.marpolbul.2009.04.025
- Ferguson, P.L., Iden, C.R., Brownawell, B.J., 2001. Distribution and Fate of Neutral Alkylphenol Ethoxylate Metabolites in a Sewage-Impacted Urban Estuary. *Environmental science & technology*. 35, 2428-2435. doi: 10.1021/es001871b

- Ferrara, F., Ademollo, N., Delise, M., Fabietti, F., Funari, E., 2008. Alkylphenols and their ethoxylates in seafood from the Tyrrhenian Sea. *Chemosphere*. 72, 1279-1285. doi: 10.1016/j.chemosphere.2008.04.060
- Foekema, E.M., De Gruijter, C., Mergia, M.T., van Franeker, J.A., Murk, A.J., Koelmans, A.A., 2013. Plastic in North sea fish. *Environmental science & technology*. 47, 8818-8824. doi: 10.1021/es400931b
- Fossi, M.C., Panti, C., Guerranti, C., Coppola, D., Giannetti, M., Marsili, L., Minutoli, R., 2012. Are baleen whales exposed to the threat of microplastics? A case study of the Mediterranean fin whale (*Balaenoptera physalus*). *Marine pollution bulletin*. 64, 2374-2379. doi: 10.1016/j.marpolbul.2012.08.013
- Fossi, M.C., Coppola, D., Baini, M., Giannetti, M., Guerranti, C., Marsili, L., Panti, C., de Sabata, E., Clò, S., 2014. Large filter feeding marine organisms as indicators of microplastic in the pelagic environment: The case studies of the Mediterranean basking shark (*Cetorhinus maximus*) and fin whale (*Balaenoptera physalus*). *Marine Environmental Research*. 100, 17-24. doi: 10.1016/j.marenvres.2014.02.002
- Fossi, M.C., Marsili, L., Baini, M., Giannetti, M., Coppola, D., Guerranti, C., Caliani, I., Minutoli, R., Lauriano, G., Finoia, M.G., Rubegni, F., Panigada, S., Bérubé, M., Urbán Ramírez, J., Panti, C., 2016. Fin whales and microplastics: The Mediterranean Sea and the Sea of Cortez scenarios. *Environmental Pollution*. 209, 68-78. doi: 10.1016/j.envpol.2015.11.022
- Fries, E., Dekiff, J.H., Willmeyer, J., Nuelle, M.-T., Ebert, M., Remy, D., 2013. Identification of polymer types and additives in marine microplastic particles using pyrolysis-GC/MS and scanning electron microscopy. *Environmental Science: Processes & Impacts*. 15, 1949-1956. doi: 10.1039/C3EM00214D
- Fu, M., Li, Z., Gao, H., 2007. Distribution characteristics of nonylphenol in Jiaozhou Bay of Qingdao and its adjacent rivers. *Chemosphere*. 69, 1009-1016. doi: 10.1016/j.chemosphere.2007.04.061
- Gandara e Silva, P.P., Nobre, C.R., Resaffe, P., Pereira, C.D.S., Gusmão, F., 2016. Leachate from microplastics impairs larval development in brown mussels. *Water Research*. 106, 364-370. doi: 10.1016/j.watres.2016.10.016
- Gassel, M., Harwani, S., Park, J.-S., Jahn, A., 2013. Detection of nonylphenol and persistent organic pollutants in fish from the North Pacific Central Gyre. *Marine pollution bulletin*. 73, 231-242. doi: 10.1016/j.marpolbul.2013.05.014
- Gaylor, M.O., Harvey, E., Hale, R.C., 2012. House crickets can accumulate polybrominated diphenyl ethers (PBDEs) directly from polyurethane foam common in consumer products. *Chemosphere*. 86, 500-505. doi: 10.1016/j.chemosphere.2011.10.014
- Giam, C., Chan, H., Neff, G., Atlas, E., 1978. Phthalate ester plasticizers: a new class of marine pollutant. *Science*. 199, 419-421. doi: 10.1126/science.413194
- Gigault, J., Pedrono, B., Maxit, B., Ter Halle, A., 2016. Marine plastic litter: the unanalyzed nano-fraction. *Environmental Science: Nano*. 3, 346-350. doi: 10.1039/C6EN00008H
- Graham, E.R., Thompson, J.T., 2009. Deposit- and suspension-feeding sea cucumbers (Echinodermata) ingest plastic fragments. *Journal of Experimental Marine Biology and Ecology*. 368, 22-29. doi: 10.1016/j.jembe.2008.09.007
- Hadjmohammadi, M.R., Fatemi, M.H., Taneh, T., 2011. Coacervative extraction of phthalates from water and their determination by high performance liquid chromatography. *Journal of the Iranian Chemical Society*. 8, 100-106. doi: 10.1007/bf03246206
- Hamlin, H.J., Marciano, K., Downs, C.A., 2015. Migration of nonylphenol from food-grade plastic is toxic to the coral reef fish species *Pseudochromis fridmani*. *Chemosphere*. 139, 223-228. doi: 10.1016/j.chemosphere.2015.06.032

- Hansen, E., Nilsson, N., Lithner, D., Lassen, C., 2013. Hazardous substances in plastic materials. Available on: <http://www.miljodirektoratet.no/old/klif/publikasjoner/3017/ta3017.pdf>, Accessed on: 11/22/2016
- Hirai, H., Takada, H., Ogata, Y., Yamashita, R., Mizukawa, K., Saha, M., Kwan, C., Moore, C., Gray, H., Laursen, D., Zettler, E.R., Farrington, J.W., Reddy, C.M., Peacock, E.E., Ward, M.W., 2011. Organic micropollutants in marine plastics debris from the open ocean and remote and urban beaches. *Marine pollution bulletin*. 62, 1683-1692. doi: 10.1016/j.marpolbul.2011.06.004
- Höfer, R., 2012. Processing and Performance Additives for Plastics. *Polymer Science: A Comprehensive Reference*, ed. K. Matyjaszewski and M. Möller. 10.
- ICIS, 2003. Product profile: Bisphenol A. Available on: <http://www.icis.com/resources/news/2003/04/24/193606/product-profile-bisphenol-a/>, Accessed on: 09/16/2016
- Industrievereinigung Chemiefaser, 2013. Man-made fibers. Available on: <https://www.ivc-ev.de/>, Accessed on: 12/22/2016
- Isobe, T., Nishiyama, H., Nakashima, A., Takada, H., 2001. Distribution and Behavior of Nonylphenol, Octylphenol, and Nonylphenol Monoethoxylate in Tokyo Metropolitan Area: Their Association with Aquatic Particles and Sedimentary Distributions. *Environmental science & technology*. 35, 1041-1049. doi: 10.1021/es001250i
- Isobe, T., Takada, H., Kanai, M., Tsutsumi, S., Isobe, K.O., Boonyatumanond, R., Zakaria, M.P., 2007. Distribution of Polycyclic Aromatic Hydrocarbons (PAHs) and phenolic endocrine disrupting chemicals in South and Southeast Asian mussels. *Environmental Monitoring and Assessment*. 135, 423-440. doi: 10.1007/s10661-007-9661-y
- Jang, M., Shim, W.J., Han, G.M., Rani, M., Song, Y.K., Hong, S.H., 2016. Styrofoam Debris as a Source of Hazardous Additives for Marine Organisms. *Environmental science & technology*. 50, 4951-4960. doi: 10.1021/acs.est.5b05485
- Johansson, I., Héas-Moisán, K., Guiot, N., Munschy, C., Tronczyński, J., 2006. Polybrominated diphenyl ethers (PBDEs) in mussels from selected French coastal sites: 1981–2003. *Chemosphere*. 64, 296-305. doi: 10.1016/j.chemosphere.2005.12.014
- Jonkers, N., Laane, R.W.P.M., de Voogt, P., 2003. Fate of Nonylphenol Ethoxylates and Their Metabolites in Two Dutch Estuaries: Evidence of Biodegradation in the Field. *Environmental science & technology*. 37, 321-327. doi: 10.1021/es020121u
- Kannan, N., Yamashita, N., Petrick, G., Duinker, J.C., 1998. Polychlorinated Biphenyls and Nonylphenols in the Sea of Japan. *Environmental science & technology*. 32, 1747-1753. doi: 10.1021/es970713q
- Kawahata, H., Ohta, H., Inoue, M., Suzuki, A., 2004. Endocrine disrupter nonylphenol and bisphenol A contamination in Okinawa and Ishigaki Islands, Japan—within coral reefs and adjacent river mouths. *Chemosphere*. 55, 1519-1527. doi: 10.1016/j.chemosphere.2004.01.032
- Keil, R., Salemme, K., Forrest, B., Neibauer, J., Logsdon, M., 2011. Differential presence of anthropogenic compounds dissolved in the marine waters of Puget Sound, WA and Barkley Sound, BC. *Marine pollution bulletin*. 62, 2404-2411. doi: 10.1016/j.marpolbul.2011.08.029
- Klamer, H.J.C., Leonards, P.E.G., Lamoree, M.H., Villerius, L.A., Åkerman, J.E., Bakker, J.F., 2005. A chemical and toxicological profile of Dutch North Sea surface sediments. *Chemosphere*. 58, 1579-1587. doi: 10.1016/j.chemosphere.2004.11.027
- Koelmans, A.A., Besseling, E., Shim, W.J., 2015. Nanoplastics in the Aquatic Environment. *Critical Review in Bergmann, M., Gutow, L., Klages, M. (Eds.), Marine Anthropogenic Litter*. Springer International Publishing, Cham, pp. 325-340.

- 824 Koelmans, A.A., Besseling, E., Wegner, A., Foekema, E.M., 2013a. Correction to Plastic As a
825 Carrier of POPs to Aquatic Organisms: A Model Analysis. *Environmental science &*
826 *technology*. 47, 8992-8993. doi: 10.1021/es403018h
- 827 Koelmans, A.A., Besseling, E., Wegner, A., Foekema, E.M., 2013b. Plastic as a Carrier of POPs
828 to Aquatic Organisms: A Model Analysis. *Environmental science & technology*. 47,
829 7812-7820. doi: 10.1021/es401169n
- 830 Koelmans, A.A., Bakir, A., Burton, G.A., Janssen, C.R., 2016. Microplastic as a Vector for
831 Chemicals in the Aquatic Environment: Critical Review and Model-Supported
832 Reinterpretation of Empirical Studies. *Environmental science & technology*. 50, 3315-
833 3326. doi: 10.1021/acs.est.5b06069
- 834 Kurihara, R., Watanabe, E., Ueda, Y., Kakuno, A., Fujii, K., Shiraishi, F., Hashimoto, S., 2007.
835 Estrogenic activity in sediments contaminated by nonylphenol in Tokyo Bay (Japan)
836 evaluated by vitellogenin induction in male mummichogs (*Fundulus heteroclitus*).
837 *Marine pollution bulletin*. 54, 1315-1320. doi: 10.1016/j.marpolbul.2007.06.007
- 838 Laing, L.V., Viana, J., Dempster, E.L., Trznadel, M., Trunkfield, L.A., Uren Webster, T.M.,
839 van Aerle, R., Paull, G.C., Wilson, R.J., Mill, J., Santos, E.M., 2016. Bisphenol A causes
840 reproductive toxicity, decreases dnmt1 transcription, and reduces global DNA
841 methylation in breeding zebrafish (*Danio rerio*). *Epigenetics*. 11, 526-538. doi:
842 10.1080/15592294.2016.1182272
- 843 Lambert, S., Wagner, M., 2016. Characterisation of nanoplastics during the degradation of
844 polystyrene. *Chemosphere*. 145, 265-268. doi: 10.1016/j.chemosphere.2015.11.078
- 845 Lau, O.-W., Wong, S.-K., 2000. Contamination in food from packaging material. *Journal of*
846 *Chromatography A*. 882, 255-270. doi: 10.1016/S0021-9673(00)00356-3
- 847 Law, R., Fileman, T., Matthiessen, P., 1991. Phthalate esters and other industrial organic
848 chemicals in the North and Irish Seas. *Water Science and Technology*. 24, 127-134.
- 849 Li, D., Dong, M., Shim, W.J., Yim, U.H., Hong, S.H., Kannan, N., 2008. Distribution
850 characteristics of nonylphenolic chemicals in Masan Bay environments, Korea.
851 *Chemosphere*. 71, 1162-1172. doi: 10.1016/j.chemosphere.2007.10.023
- 852 Li, H.-X., Getzinger, G.J., Ferguson, P.L., Orihuela, B., Zhu, M., Rittschof, D., 2016. Effects
853 of Toxic Leachate from Commercial Plastics on Larval Survival and Settlement of the
854 Barnacle *Amphibalanus amphitrite*. *Environmental science & technology*. 50, 924-931.
855 doi: 10.1021/acs.est.5b02781
- 856 Lithner, D., Larsson, Å., Dave, G., 2011. Environmental and health hazard ranking and
857 assessment of plastic polymers based on chemical composition. *Science of The Total*
858 *Environment*. 409, 3309-3324. doi: 10.1016/j.scitotenv.2011.04.038
- 859 Lithner, D., Nordensvan, I., Dave, G., 2012. Comparative acute toxicity of leachates from
860 plastic products made of polypropylene, polyethylene, PVC, acrylonitrile–butadiene–
861 styrene, and epoxy to *Daphnia magna*. *Environmental Science and Pollution Research*.
862 19, 1763-1772. doi: 10.1007/s11356-011-0663-5
- 863 Lithner, D., Damberg, J., Dave, G., Larsson, Å., 2009. Leachates from plastic consumer
864 products – Screening for toxicity with *Daphnia magna*. *Chemosphere*. 74, 1195-1200.
865 doi: 10.1016/j.chemosphere.2008.11.022
- 866 Loyo-Rosales, J.E., Rosales-Rivera, G.C., Lynch, A.M., Rice, C.P., Torrents, A., 2004.
867 Migration of Nonylphenol from Plastic Containers to Water and a Milk Surrogate.
868 *Journal of Agricultural and Food Chemistry*. 52, 2016-2020. doi: 10.1021/jf0345696
- 869 Lusher, A., McHugh, M., Thompson, R., 2013. Occurrence of microplastics in the
870 gastrointestinal tract of pelagic and demersal fish from the English Channel. *Marine*
871 *pollution bulletin*. 67, 94-99. doi: 10.1016/j.marpolbul.2012.11.028
- 872 Lusher, A.L., Hernandez-Milian, G., O'Brien, J., Berrow, S., O'Connor, I., Officer, R., 2015.
873 Microplastic and macroplastic ingestion by a deep diving, oceanic cetacean: The True's

- 874 beaked whale *Mesoplodon mirus*. *Environmental Pollution*. 199, 185-191. doi:
875 10.1016/j.envpol.2015.01.023
- 876 Mackintosh, C.E., Maldonado, J., Hongwu, J., Hoover, N., Chong, A., Ikonomou, M.G., Gobas,
877 F.A.P.C., 2004. Distribution of Phthalate Esters in a Marine Aquatic Food Web:
878 Comparison to Polychlorinated Biphenyls. *Environmental science & technology*. 38,
879 2011-2020. doi: 10.1021/es034745r
- 880 Mato, Y., Isobe, T., Takada, H., Kanehiro, H., Ohtake, C., Kaminuma, T., 2001. Plastic Resin
881 Pellets as a Transport Medium for Toxic Chemicals in the Marine Environment.
882 *Environmental science & technology*. 35, 318-324. doi: 10.1021/es0010498
- 883 Meeker, J.D., Sathyanarayana, S., Swan, S.H., 2009. Phthalates and other additives in plastics:
884 human exposure and associated health outcomes. *Philosophical Transactions of the*
885 *Royal Society B: Biological Sciences*. 364, 2097-2113. doi: 10.1098/rstb.2008.0268
- 886 Moon, H.-B., Kannan, K., Choi, M., Choi, H.-G., 2007a. Polybrominated diphenyl ethers
887 (PBDEs) in marine sediments from industrialized bays of Korea. *Marine pollution*
888 *bulletin*. 54, 1402-1412. doi: 10.1016/j.marpolbul.2007.05.024
- 889 Moon, H.-B., Kannan, K., Lee, S.-J., Choi, M., 2007b. Polybrominated diphenyl ethers
890 (PBDEs) in sediment and bivalves from Korean coastal waters. *Chemosphere*. 66, 243-
891 251. doi: 10.1016/j.chemosphere.2006.05.025
- 892 Morris, S., Allchin, C.R., Zegers, B.N., Haftka, J.J.H., Boon, J.P., Belpaire, C., Leonards,
893 P.E.G., van Leeuwen, S.P.J., de Boer, J., 2004. Distribution and Fate of HBCD and
894 TBBPA Brominated Flame Retardants in North Sea Estuaries and Aquatic Food Webs.
895 *Environmental science & technology*. 38, 5497-5504. doi: 10.1021/es049640i
- 896 Napper, I.E., Bakir, A., Rowland, S.J., Thompson, R.C., 2015. Characterisation, quantity and
897 sorptive properties of microplastics extracted from cosmetics. *Marine pollution bulletin*.
898 99, 178-185. doi: 10.1016/j.marpolbul.2015.07.029
- 899 Net, S., Sempéré, R., Delmont, A., Paluselli, A., Ouddane, B., 2015. Occurrence, Fate, Behavior
900 and Ecotoxicological State of Phthalates in Different Environmental Matrices.
901 *Environmental science & technology*. 49, 4019-4035. doi: 10.1021/es505233b
- 902 Neves, D., Sobral, P., Ferreira, J.L., Pereira, T., 2015. Ingestion of microplastics by commercial
903 fish off the Portuguese coast. *Marine pollution bulletin*. 101, 119-126. doi:
904 10.1016/j.marpolbul.2015.11.008
- 905 OECD, 2004. Emission scenario document on plastic additives. Series on emission scenario
906 documents, no. 3. Paris: Environmental directorate, OECD Environmental Health and
907 Safety Publications. Available on:
908 [http://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?doclanguage=en&](http://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?doclanguage=en&cote=ENV/JM/MONO%282004%298/REV1)
909 [cote=ENV/JM/MONO%282004%298/REV1](http://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?doclanguage=en&cote=ENV/JM/MONO%282004%298/REV1), Accessed on: 06/15/2016
- 910 Oehlmann, J., Oetken, M., Schulte-Oehlmann, U., 2008. A critical evaluation of the
911 environmental risk assessment for plasticizers in the freshwater environment in Europe,
912 with special emphasis on bisphenol A and endocrine disruption. *Environmental*
913 *Research*. 108, 140-149. doi: 10.1016/j.envres.2008.07.016
- 914 Oehlmann, J., Schulte-Oehlmann, U., Kloas, W., Jagnytsch, O., Lutz, I., Kusk, K.O.,
915 Wollenberger, L., Santos, E.M., Paull, G.C., Van Look, K.J.W., Tyler, C.R., 2009. A
916 critical analysis of the biological impacts of plasticizers on wildlife. *Philosophical*
917 *Transactions of the Royal Society of London B: Biological Sciences*. 364, 2047-2062.
918 doi: 10.1098/rstb.2008.0242
- 919 Paul-Pont, I., Lacroix, C., González Fernández, C., Hégaret, H., Lambert, C., Le Goïc, N.,
920 Frère, L., Cassone, A.-L., Sussarellu, R., Fabioux, C., Guyomarch, J., Albentosa, M.,
921 Huvet, A., Soudant, P., 2016. Exposure of marine mussels *Mytilus* spp. to polystyrene
922 microplastics: Toxicity and influence on fluoranthene bioaccumulation. *Environmental*
923 *Pollution*. 216, 724-737. doi: 10.1016/j.envpol.2016.06.039

- 924 Peng, J.-H., Huang, C.-W., Weng, Y.-M., Yak, H.-K., 2007. Determination of polybrominated
925 diphenyl ethers (PBDEs) in fish samples from rivers and estuaries in Taiwan.
926 *Chemosphere*. 66, 1990-1997. doi: 10.1016/j.chemosphere.2006.07.094
- 927 Petrovic, M., Fernández-Alba, A.R., Borrull, F., Marce, R.M., Mazo, E.G., Barceló, D., 2002.
928 Occurrence and distribution of nonionic surfactants, their degradation products, and
929 linear alkylbenzene sulfonates in coastal waters and sediments in Spain. *Environmental*
930 *Toxicology and Chemistry*. 21, 37-46. doi: 10.1002/etc.5620210106
- 931 PlasticsEurope, 2016. Plastics – the Facts 2016: An analysis of European plastics production,
932 demand and waste data. Available on: [http://www.plasticseurope.fr/Document/plastics-](http://www.plasticseurope.fr/Document/plastics--the-facts-2016-15787.aspx?FolID=2)
933 [--the-facts-2016-15787.aspx?FolID=2](http://www.plasticseurope.fr/Document/plastics--the-facts-2016-15787.aspx?FolID=2), Accessed on: 10/24/2016
- 934 Pojana, G., Gomiero, A., Jonkers, N., Marcomini, A., 2007. Natural and synthetic endocrine
935 disrupting compounds (EDCs) in water, sediment and biota of a coastal lagoon.
936 *Environment International*. 33, 929-936. doi: 10.1016/j.envint.2007.05.003
- 937 Possatto, F.E., Barletta, M., Costa, M.F., Ivar do Sul, J.A., Dantas, D.V., 2011. Plastic debris
938 ingestion by marine catfish: An unexpected fisheries impact. *Marine pollution bulletin*.
939 62, 1098-1102. doi: 10.1016/j.marpolbul.2011.01.036
- 940 Prieto, A., Zuloaga, O., Usobiaga, A., Etxebarria, N., Fernández, L.A., 2007. Development of
941 a stir bar sorptive extraction and thermal desorption–gas chromatography–mass
942 spectrometry method for the simultaneous determination of several persistent organic
943 pollutants in water samples. *Journal of Chromatography A*. 1174, 40-49. doi:
944 10.1016/j.chroma.2007.07.054
- 945 Ramu, K., Kajiwarra, N., Isobe, T., Takahashi, S., Kim, E.-Y., Min, B.-Y., We, S.-U., Tanabe,
946 S., 2007. Spatial distribution and accumulation of brominated flame retardants,
947 polychlorinated biphenyls and organochlorine pesticides in blue mussels (*Mytilus*
948 *edulis*) from coastal waters of Korea. *Environmental Pollution*. 148, 562-569. doi:
949 10.1016/j.envpol.2006.11.034
- 950 Rani, M., Shim, W.J., Han, G.M., Jang, M., Al-Odaini, N.A., Song, Y.K., Hong, S.H., 2015.
951 Qualitative Analysis of Additives in Plastic Marine Debris and Its New Products.
952 *Archives of Environmental Contamination and Toxicology*. 69, 352-366. doi:
953 10.1007/s00244-015-0224-x
- 954 Rochman, C.M., Hoh, E., Kurobe, T., Teh, S.J., 2013. Ingested plastic transfers hazardous
955 chemicals to fish and induces hepatic stress. *Scientific reports*. 3, 3263. doi:
956 10.1038/srep03263
- 957 Rochman, C.M., Lewison, R.L., Eriksen, M., Allen, H., Cook, A.-M., Teh, S.J., 2014.
958 Polybrominated diphenyl ethers (PBDEs) in fish tissue may be an indicator of plastic
959 contamination in marine habitats. *Science of The Total Environment*. 476–477, 622-
960 633. doi: 10.1016/j.scitotenv.2014.01.058
- 961 Rochman, C.M., Tahir, A., Williams, S.L., Baxa, D.V., Lam, R., Miller, J.T., Teh, F.-C.,
962 Werorilangi, S., Teh, S.J., 2015. Anthropogenic debris in seafood: Plastic debris and
963 fibers from textiles in fish and bivalves sold for human consumption. *Scientific reports*.
964 5. doi: 10.1038/srep14340
- 965 Sajiki, J., Yonekubo, J., 2003. Leaching of bisphenol A (BPA) to seawater from polycarbonate
966 plastic and its degradation by reactive oxygen species. *Chemosphere*. 51, 55-62. doi:
967 10.1016/S0045-6535(02)00789-0
- 968 Sánchez-Avila, J., Tauler, R., Lacorte, S., 2012. Organic micropollutants in coastal waters from
969 NW Mediterranean Sea: Sources distribution and potential risk. *Environment*
970 *International*. 46, 50-62. doi: 10.1016/j.envint.2012.04.013
- 971 Schlenk, D., Sapozhnikova, Y., Irwin, M.A., Xie, L., Hwang, W., Reddy, S., Brownawell, B.J.,
972 Armstrong, J., Kelly, M., Montagne, D.E., Kolodziej, E.P., Sedlak, D., Snyder, S., 2005.
973 In vivo bioassay-guided fractionation of marine sediment extracts from the Southern

- California Bight, USA, for estrogenic activity. *Environmental Toxicology and Chemistry*. 24, 2820-2826. doi: 10.1897/05-116R.1
- Soares, A., Guieysse, B., Jefferson, B., Cartmell, E., Lester, J.N., 2008. Nonylphenol in the environment: A critical review on occurrence, fate, toxicity and treatment in wastewaters. *Environment International*. 34, 1033-1049. doi: 10.1016/j.envint.2008.01.004
- Stockholm Convention, 2016. Listing of POPs in the Stockholm Convention. Available on: <http://chm.pops.int/TheConvention/ThePOPs/ListingofPOPs/tabid/2509/Default.aspx>, Accessed on: 09/12/2016
- Sussarellu, R., Suquet, M., Thomas, Y., Lambert, C., Fabioux, C., Pernet, M.E.J., Le Goïc, N., Quillien, V., Mingant, C., Epelboin, Y., Corporeau, C., Guyomarch, J., Robbins, J., Paul-Pont, I., Soudant, P., Huvet, A., 2016. Oyster reproduction is affected by exposure to polystyrene microplastics. *Proceedings of the National Academy of Sciences*. 113, 2430-2435. doi: 10.1073/pnas.1519019113
- Talsness, C.E., Andrade, A.J.M., Kuriyama, S.N., Taylor, J.A., vom Saal, F.S., 2009. Components of plastic: experimental studies in animals and relevance for human health. *Philosophical Transactions of the Royal Society B: Biological Sciences*. 364, 2079-2096. doi: 10.1098/rstb.2008.0281
- Tan, G.H., 1995. Residue levels of phthalate esters in water and sediment samples from the Klang River basin. *Bulletin of Environmental Contamination and Toxicology*. 54, 171-176. doi: 10.1007/bf00197427
- Tanaka, K., Takada, H., Yamashita, R., Mizukawa, K., Fukuwaka, M.-a., Watanuki, Y., 2013. Accumulation of plastic-derived chemicals in tissues of seabirds ingesting marine plastics. *Marine pollution bulletin*. 69, 219-222. doi: 10.1016/j.marpolbul.2012.12.010
- Tanaka, K., Takada, H., Yamashita, R., Mizukawa, K., Fukuwaka, M.-a., Watanuki, Y., 2015. Facilitated Leaching of Additive-Derived PBDEs from Plastic by Seabirds' Stomach Oil and Accumulation in Tissues. *Environmental science & technology*. 49, 11799-11807. doi: 10.1021/acs.est.5b01376
- Teuten, E.L., Rowland, S.J., Galloway, T.S., Thompson, R.C., 2007. Potential for Plastics to Transport Hydrophobic Contaminants. *Environmental science & technology*. 41, 7759-7764. doi: 10.1021/es071737s
- Teuten, E.L., Saquing, J.M., Knappe, D.R.U., Barlaz, M.A., Jonsson, S., Björn, A., Rowland, S.J., Thompson, R.C., Galloway, T.S., Yamashita, R., Ochi, D., Watanuki, Y., Moore, C., Viet, P.H., Tana, T.S., Prudente, M., Boonyatumanond, R., Zakaria, M.P., Akkhavong, K., Ogata, Y., Hirai, H., Iwasa, S., Mizukawa, K., Hagino, Y., Imamura, A., Saha, M., Takada, H., 2009. Transport and release of chemicals from plastics to the environment and to wildlife. *Philosophical Transactions of the Royal Society of London B: Biological Sciences*. 364, 2027-2045. doi: 10.1098/rstb.2008.0284
- Thompson, R.C., Swan, S.H., Moore, C.J., vom Saal, F.S., 2009. Our plastic age. *Philosophical Transactions of the Royal Society B: Biological Sciences*. 364, 1973-1976. doi: 10.1098/rstb.2009.0054
- Thompson, R.C., Olsen, Y., Mitchell, R.P., Davis, A., Rowland, S.J., John, A.W., McGonigle, D., Russell, A.E., 2004. Lost at sea: where is all the plastic? *Science*. 304, 838-838. doi: 10.1126/science.1094559
- US Environmental Protection Agency, 2006. Certain Polybrominated Diphenylethers; Significant New Use Rule. Available on: <https://www.federalregister.gov/documents/2006/06/13/E6-9207/certain-polybrominated-diphenylethers-significant-new-use-rule>, Accessed on: 04/26/2017
- US Environmental Protection Agency, 2010a. Nonylphenol (NP) and Nonylphenol Ethoxylates (NPEs) Action Plan. Available on: <https://www.epa.gov/sites/production/files/2015->

- 1024 [09/documents/rin2070-za09_np-npes_action_plan_final_2010-08-09.pdf](#), Accessed
1025 on: 05/04/2016
- 1026 US Environmental Protection Agency, 2010b. Hexabromocyclododecane (HBCD) Action Plan.
1027 Available on: [https://www.epa.gov/assessing-and-managing-chemicals-under-](https://www.epa.gov/assessing-and-managing-chemicals-under-tsca/hexabromocyclododecane-hbcd-action-plan)
1028 [tsca/hexabromocyclododecane-hbcd-action-plan](https://www.epa.gov/assessing-and-managing-chemicals-under-tsca/hexabromocyclododecane-hbcd-action-plan), Accessed on: 04/26/2017
- 1029 US Environmental Protection Agency, 2012. Certain Polybrominated Diphenylethers;
1030 Significant New Use Rule and Test Rule. Available on:
1031 [http://blogs.edf.org/health/files/2012/07/EDF_Earthjustice-Comments-on-Proposed-](http://blogs.edf.org/health/files/2012/07/EDF_Earthjustice-Comments-on-Proposed-PBDE-test-rule-and-SNUR-FINAL-7-31-12.pdf)
1032 [PBDE-test-rule-and-SNUR-FINAL-7-31-12.pdf](http://blogs.edf.org/health/files/2012/07/EDF_Earthjustice-Comments-on-Proposed-PBDE-test-rule-and-SNUR-FINAL-7-31-12.pdf), Accessed on: 04/26/2017
- 1033 Van Cauwenberghe, L., Janssen, C.R., 2014. Microplastics in bivalves cultured for human
1034 consumption. *Environmental Pollution*. 193, 65-70. doi: 10.1016/j.envpol.2014.06.010
- 1035 Van Cauwenberghe, L., Claessens, M., Vandegehuchte, M.B., Janssen, C.R., 2015.
1036 Microplastics are taken up by mussels (*Mytilus edulis*) and lugworms (*Arenicola*
1037 *marina*) living in natural habitats. *Environmental Pollution*. 199, 10-17. doi:
1038 10.1016/j.envpol.2015.01.008
- 1039 Van Sebillie, E., Wilcox, C., Lebreton, L., Maximenko, N., Hardesty, B.D., Van Franeker, J.A.,
1040 Eriksen, M., Siegel, D., Galgani, F., Law, K.L., 2015. A global inventory of small
1041 floating plastic debris. *Environmental Research Letters*. 10, 124006. doi: 10.1088/1748-
1042 9326/10/12/124006
- 1043 Vandermeersch, G., Lourenço, H.M., Alvarez-Muñoz, D., Cunha, S., Diogène, J., Cano-
1044 Sancho, G., Sloth, J.J., Kwadijk, C., Barcelo, D., Allegaert, W., Bekaert, K., Fernandes,
1045 J.O., Marques, A., Robbins, J., 2015. Environmental contaminants of emerging concern
1046 in seafood – European database on contaminant levels. *Environmental Research*. 143,
1047 Part B, 29-45. doi: 10.1016/j.envres.2015.06.011
- 1048 Verslycke, T.A., Vethaak, A.D., Arijs, K., Janssen, C.R., 2005. Flame retardants, surfactants
1049 and organotins in sediment and mysid shrimp of the Scheldt estuary (The Netherlands).
1050 *Environmental Pollution*. 136, 19-31. doi: 10.1016/j.envpol.2004.12.008
- 1051 Vethaak, A.D., Lahr, J., Schrap, S.M., Belfroid, A.C., Rijs, G.B.J., Gerritsen, A., de Boer, J.,
1052 Bulder, A.S., Grinwis, G.C.M., Kuiper, R.V., Legler, J., Murk, T.A.J., Peijnenburg, W.,
1053 Verhaar, H.J.M., de Voogt, P., 2005. An integrated assessment of estrogenic
1054 contamination and biological effects in the aquatic environment of The Netherlands.
1055 *Chemosphere*. 59, 511-524. doi: 10.1016/j.chemosphere.2004.12.053
- 1056 Wardrop, P., Shimeta, J., Nugegoda, D., Morrison, P.D., Miranda, A., Tang, M., Clarke, B.O.,
1057 2016. Chemical Pollutants Sorbed to Ingested Microbeads from Personal Care Products
1058 Accumulate in Fish. *Environmental science & technology*. 50, 4037-4044. doi:
1059 10.1021/acs.est.5b06280
- 1060 Wright, S.L., Thompson, R.C., Galloway, T.S., 2013. The physical impacts of microplastics on
1061 marine organisms: a review. *Environmental Pollution*. 178, 483-492. doi:
1062 10.1016/j.envpol.2013.02.031
- 1063 Wurl, O., Lam, P.K.S., Obbard, J.P., 2006. Occurrence and distribution of polybrominated
1064 diphenyl ethers (PBDEs) in the dissolved and suspended phases of the sea-surface
1065 microlayer and seawater in Hong Kong, China. *Chemosphere*. 65, 1660-1666. doi:
1066 10.1016/j.chemosphere.2006.02.024
- 1067 Xie, Z., Ebinghaus, R., Temme, C., Caba, A., Ruck, W., 2005. Atmospheric concentrations and
1068 air-sea exchanges of phthalates in the North Sea (German Bight). *Atmospheric*
1069 *Environment*. 39, 3209-3219. doi: 10.1016/j.atmosenv.2005.02.021
- 1070 Xie, Z., Lakaschus, S., Ebinghaus, R., Caba, A., Ruck, W., 2006. Atmospheric concentrations
1071 and air-sea exchanges of nonylphenol, tertiary octylphenol and nonylphenol
1072 monoethoxylate in the North Sea. *Environmental Pollution*. 142, 170-180. doi:
1073 10.1016/j.envpol.2005.08.073

- 1074 Xie, Z., Ebinghaus, R., Temme, C., Lohmann, R., Caba, A., Ruck, W., 2007. Occurrence and
1075 Air–Sea Exchange of Phthalates in the Arctic. *Environmental science & technology*. 41,
1076 4555-4560. doi: 10.1021/es0630240
1077 Zitko, V., Hanlon, M., 1991. Another source of pollution by plastics: skin cleaners with plastic
1078 scrubbers. *Marine pollution bulletin*. 22, 41-42.

1079